

A Project report on
IMPROVED PROTECTION AND MONITORING OF
BATTERY IN ELECTRIC VEHICLES USING IoT

Submitted in partial fulfilment of the requirements for the award of the Degree of

BACHELOR OF TECHNOLOGY
IN
ELECTRICAL AND ELECTRONICS ENGINEERING

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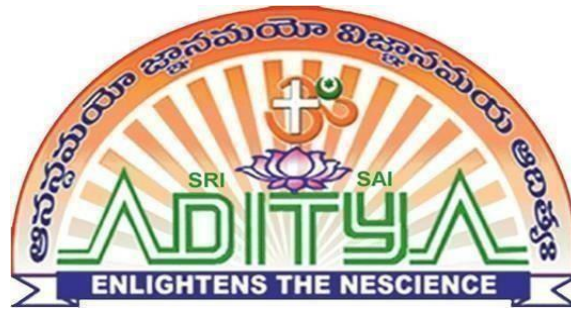


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We are thankful to our guide **Mr. RAJESH MURARI**, M.Tech.,(Ph. D), spared his valuable time append novel ideas to guide us in intelligent. We are indebted to him without whom we not have culminated to pinnacle of the project.

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DECLARATION

We are here by declare that this project titled “**“IMPROVED PROTECTION AND MONITORING OF BATTERY IN ELECTRIC VEHICLES USING IoT”**” has been under taken at this work has been submitted to **ADITYA COLLEGE OF ENGINEERING AND TECHNOLOGY**, Surampalem, affiliated to **JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, Kakinada** in partial fulfillment of the Degree of Bachelors of Technology.

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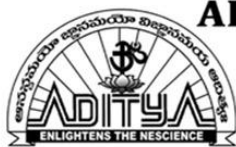
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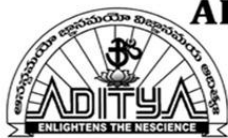
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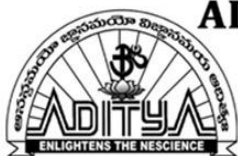
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ABSTRACT

The conventional energy supplied by the electric utility is costly, and effects environment, hence customers are turning to distributed power sources (DPSs) like solar energy for many applications. The charging infrastructure are integrated with DPSs for charging batteries inside electric vehicles (EVs). However, these batteries in EVs are burning or blasting due to the overcharging/ long time charging due to excruciated temperature rise inside battery. Sometimes this may happen due to over voltage. In order to avoid these mishaps, in this project we propose a special type of battery charger for electric vehicles or bikes. The proposed charger is designed using IoT based monitoring and protection circuit. The battery temperature is continuously monitored through an Arduino based embedded system. The supply to the battery will be disconnected if the temperature reaches a preset threshold value, and will be restored once the reset button is activated. Additionally, information can be monitored through the data transmitted to connected smartphone or device. The proposed method enhances protection, efficiency and life cycle of the battery in EVs.

CHAPTER 1

CHAPTER 1

INTRODUCTION

Electric vehicles (EVs) are transforming the landscape of modern transportation by offering an eco-friendly alternative to conventional fossil fuel-powered vehicles. As the world confronts pressing challenges related to climate change, environmental pollution, and the depletion of non-renewable energy sources, the adoption of EVs presents a sustainable and promising path forward. However, the success of EVs hinges largely on two crucial factors — the efficiency of the battery charging infrastructure and the safety mechanisms involved in battery management and protection.

This project aims to address both aspects by developing a solar-based 48V e-bike charger integrated with Internet of Things (IoT) capabilities to enhance monitoring and protection. The solution leverages renewable solar energy for charging while ensuring the battery is safeguarded against hazards like overheating. It provides real-time data on battery status through IoT sensors and remote monitoring, promoting a safer and more intelligent EV ecosystem.

The following subsections explore the motivation behind this project, the importance of battery protection systems in EVs, the role of IoT, the shift from conventional charging to solar-based alternatives, and how this integrated solution can contribute to the growth of sustainable electric mobility.

1.1 Motivation for the Project

The rapid urbanization and rise in population density have significantly increased the demand for clean, efficient, and cost-effective transportation systems. Traditional vehicles running on petrol or diesel not only add to greenhouse gas emissions but also rely on finite fossil fuel resources. As a response to these challenges, electric vehicles have emerged as a sustainable transportation alternative. However, the charging infrastructure for EVs, especially two-wheelers like e-bikes, remains a major concern in many parts of the world, including rural and semi-urban areas. Another concern is battery safety. Without adequate protection mechanisms, batteries are susceptible to thermal runaway, voltage imbalance, and capacity degradation. Addressing these risks through an intelligent, renewable-powered charging system motivated the development of this project. This project seeks to combine three key technologies — solar energy, electric vehicles, and IoT — into a single solution that not only charges EVs using clean energy but also safeguards battery health through smart protection features. By integrating IoT sensors, users can monitor key parameters such as battery temperature, voltage levels, and

charging status in real-time, reducing the chances of accidents and improving user confidence in EV technology.

1.2 Importance of EV Battery Protection Systems

The battery is arguably the most critical and expensive component of an electric vehicle. Its safety, performance, and lifecycle directly impact the overall functionality and reliability of the vehicle. As EV adoption increases, ensuring battery health and safety becomes essential. Lithium-ion batteries, which are widely used in EVs due to their high energy density and performance, are particularly sensitive to environmental conditions and charging behaviour.

EV batteries face several operational risks including overheating, short-circuits, and deep discharging. Any of these conditions can lead to catastrophic failures, posing safety hazards for users and reducing the battery's lifespan. Therefore, a battery protection system is indispensable for preventing such issues. A robust protection system constantly monitors voltage and temperature across battery cells. It can disconnect the load or stop charging when thresholds are exceeded. Furthermore, the inclusion of smart technology like IoT enhances this protection by allowing real-time data logging, alerts, and remote control, enabling proactive safety measures and maintenance planning. In the proposed system, sensors are used to detect and relay battery health data, while microcontrollers and IoT platforms manage alerts and decision-making. This level of intelligence is critical not only for individual safety but also for building trust in EV technologies, ultimately supporting their mass adoption.



Fig.1.1 EV Bike Battery Problem

1.3 Role of IoT in Smart EV Charging and Protection

The Internet of Things (IoT) plays a revolutionary role in transforming conventional charging systems into intelligent, self-regulating platforms. With the rise of smart technologies, IoT has enabled systems to gather, transmit, and analyze real-time data from various sensors. In the

context of EV battery charging and protection, IoT provides remote monitoring, early warning mechanisms, predictive maintenance, and improved user interaction.

In this project, IoT sensors are embedded in the battery charging setup to continuously monitor critical parameters like voltage, temperature, and charging status. These values are transmitted to cloud platforms or local monitoring systems via Wi-Fi or other communication protocols. When any reading exceeds safe limits, alerts are generated instantly, and appropriate actions — such as stopping the charge or disconnecting the battery — can be automatically initiated.

IoT's integration not only enhances safety but also adds a layer of automation and intelligence, making the charging process seamless, efficient, and user-friendly. As the demand for smarter EV infrastructure increases, IoT becomes an essential pillar in shaping future-ready charging systems.

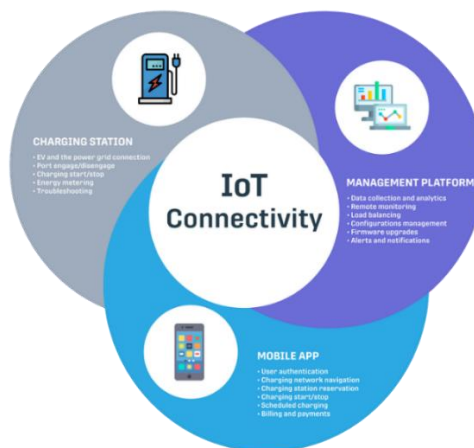


Fig.1.2 IOT Connectivity

1.4 Shift from Conventional to Solar-Based Charging

Traditional EV charging methods rely heavily on electricity from the grid, which is often generated from non-renewable sources such as coal and natural gas. While EVs themselves are eco-friendly in operation, if their electricity comes from polluting sources, the environmental benefits are offset. This realization has sparked a growing interest in renewable energy-powered charging solutions, particularly solar energy, which is abundant, clean, and sustainable. Solar-based EV charging is especially beneficial in regions with high solar insolation. It reduces dependency on grid power, lowers electricity costs, and significantly decreases the carbon footprint associated with EV use. Additionally, it provides an independent and decentralized energy source, which is valuable in remote or rural areas where grid access may be unreliable. The proposed project utilizes a solar panel setup to charge a 48V battery used in e-bikes. The power generated is regulated and managed through appropriate converters and

controllers to ensure the battery receives the correct voltage and current levels. This not only makes the system environmentally friendly but also cost-effective in the long term. Combining solar energy with smart protection and monitoring features results in a highly efficient and reliable EV charging solution. It represents a forward-thinking approach to sustainable transportation and energy management.

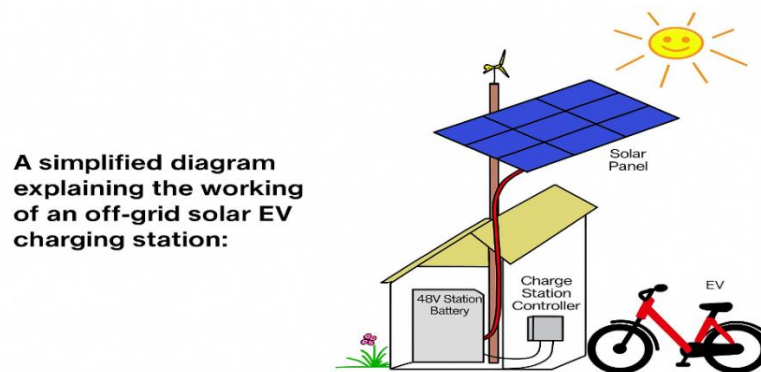


Fig.1.3 Charging Structure

1.5 Challenges in EV Charging and Battery Management

One of the primary concerns in electric vehicle (EV) battery management is **overheating during the charging process**. Unlike conventional fuel-based systems, EV batteries are highly sensitive to temperature fluctuations. When the battery temperature rises excessively—especially beyond 40°C—it can lead to serious safety hazards and long-term degradation of battery health. High temperatures during charging can trigger chemical imbalances within lithium-ion cells, which are commonly used in EVs. These imbalances may result in cell swelling, leakage, or even thermal runaway, a situation where the battery heats uncontrollably. Continuous exposure to such temperatures accelerates **battery aging**, reducing the battery's capacity and efficiency over time. Moreover, overheated cells are more prone to internal short circuits, which pose a fire risk. To mitigate this risk, our project implements a **protective disconnection mechanism** based on real-time temperature monitoring. Using IoT-enabled sensors, the system continuously tracks the battery's thermal state during the charging process. When the temperature exceeds a critical threshold of **40°C**, the system automatically disconnects the charging circuit. This immediate response prevents the battery from entering dangerous thermal conditions, reducing the risk of damage, fire, or reduced battery life. This automated disconnect feature provides a **cost-effective and reliable** way to protect the battery without human intervention. It reduces the need for continuous manual monitoring and prevents damage caused by delayed responses. By acting swiftly when high temperatures are detected, the system helps maintain optimal battery performance and prolongs its lifespan.

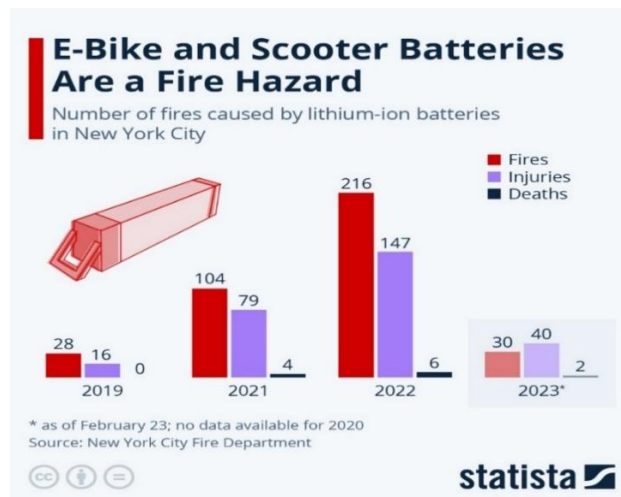


Fig.1.4 Battery Fire Hazards Statistics

1.6 Overview of the Proposed System

This project proposes a comprehensive solution that integrates solar energy, EV charging, and IoT-based battery protection to deliver an efficient and intelligent e-bike charging station. The system operates by harvesting solar energy, which is then processed and directed toward the battery through a regulated charging pathway. Meanwhile, the IoT-enabled sensors track critical battery metrics and send real-time data to a cloud dashboard or mobile app. If any unsafe condition is detected such as high temperature or voltage irregularities the system takes corrective actions and notifies the user. This approach enhances safety, improves battery life, promotes clean energy, and provides a modern, tech-savvy solution suitable for the future of electric mobility. It reflects the potential of combining multiple technologies to solve real-world problems with an eye toward innovation and sustainability.

CHAPTER 2

CHAPTER 2

Literature Review and Background Study

2.1 Introduction

The literature survey provides a foundation for understanding advancements in electric vehicle (EV) charging systems, especially those powered by renewable energy sources like solar power. With growing emphasis on sustainable transportation, solar-based EV charging has gained attention for its potential to reduce environmental impact and grid dependency.

Electric mobility, particularly in the form of e-bikes and two-wheelers, is rapidly expanding in urban and semi-urban regions. However, limited charging infrastructure, unreliable grid access, and battery degradation remain significant challenges—especially in remote areas. Solar-powered standalone systems offer a clean and practical alternative in such contexts.

Battery safety is another critical concern. Factors like overcharging and high temperatures can shorten battery life or pose safety risks. Traditional chargers often lack real-time monitoring, which has led to growing interest in integrating IoT for improved battery protection.

This chapter reviews key research in solar-based charging and IoT-enabled battery monitoring, with a focus on overtemperature detection. It identifies existing technological gaps and sets the stage for developing a safer, solar-powered charging solution for low-voltage electric bikes.

2.2 Survey of Literature

2.2.1 Solar-Based Electric Vehicle Charging Systems

Ghosh et al. [1] – MPPT-Based Solar EV Charging System

Ghosh et al. presented a detailed study on the development of a solar-powered EV charging system integrated with MPPT (Maximum Power Point Tracking) controllers. The system aimed to improve the overall efficiency of photovoltaic (PV) energy conversion under varying sunlight conditions. Their model used adaptive tracking algorithms to dynamically adjust the PV panel's operating point to the maximum power point. This ensured optimal energy transfer throughout the day, despite environmental fluctuations. They tested their setup with different battery chemistries and found improved charging consistency. Additionally, the study emphasized environmental benefits such as carbon footprint reduction and reduced reliance on grid

electricity.

Challoob and Rahmat [2] – Standalone Solar Charging for Remote Areas

Challoob and Rahmat investigated the applicability of standalone solar charging systems for electric vehicles in remote and off-grid locations. The study focused on reducing the dependency on the power grid and emphasized energy autonomy. They proposed a modular design consisting of PV panels, a battery storage unit, and a charge controller capable of handling intermittent solar inputs. The system was designed to function under harsh environmental conditions with minimal maintenance requirements. The authors argued that solar-based chargers are particularly beneficial for developing countries, where grid access is limited. Their model also allowed for easy scalability based on the charging demand. One of the key contributions was the inclusion of local energy storage to support nighttime or cloudy-day operations. The study also addressed cost factors and lifecycle analysis, suggesting long-term savings over diesel or grid-based alternatives.

Kumar et al. [3] – Cost-Effective Solar Charging for E-Bikes

Kumar et al. conducted research focusing on the design and implementation of a cost-effective solar charging station tailored for electric two-wheelers, such as e-bikes. Their setup included basic photovoltaic modules connected directly to a DC battery through a simple charge controller. The aim was to keep the system affordable while ensuring essential protection features such as overcharge and undervoltage cutoffs. The authors found that such a minimalistic system worked well for low-power applications, particularly in residential or institutional settings. They also highlighted the system's portability, making it suitable for on-the-go or mobile charging units. Additionally, they performed a cost analysis comparing grid and solar charging methods, with solar proving to be more economical in the long term. The study also incorporated basic user interface features like charge status indicators.

Rathore et al. [4] – 48V Regulated Solar Charger for Lightweight EVs

Rathore et al. focused on the development of a 48V solar charger specifically designed for lightweight electric vehicles such as scooters and e-bikes. The design featured a regulated DC output using feedback-controlled voltage regulators to maintain a constant charging voltage. The paper emphasized the importance of voltage stability in enhancing battery lifespan and performance. Their system was tested under varying solar irradiance and load conditions to ensure robustness. They also implemented a thermal sensor to protect against overheating during charging cycles. The authors discussed the use of PWM-based regulation for efficient voltage

control. Their setup also included surge protection features to safeguard sensitive electronics within the EV. The results showed stable output with minimal ripple and noise. Additionally, the study evaluated the efficiency of the system when subjected to different shading and load profiles. The authors concluded that a well-regulated 48V system can significantly enhance battery safety and reliability in solar-based chargers.

Mishra et al. [5] – IoT-Based Solar EV Charger with Monitoring

Mishra et al. developed an innovative solar-powered EV charger integrated with Internet of Things (IoT) features to monitor critical charging parameters. Their design focused on tracking temperature and voltage levels in real time to prevent overtemperature and overvoltage conditions. The IoT module sent data to a cloud dashboard, providing alerts and logs to the user or operator. The charging circuit was built around a DC supply regulated via microcontroller-based logic. One key strength of this system was the proactive safety management it offered through remote monitoring. The authors also included a mobile app for end-user interaction, enabling convenient system status checks. Additionally, the system performed automatic cut-off when abnormal values were detected. The research demonstrated that the integration of IoT not only enhances user experience but also significantly improves battery protection

2.2.2 Challenges in Grid-Based EV Charging Systems

Yadav et al. [6] – Grid Overloading and Power Quality Concerns

Yadav et al. addressed a critical issue faced by modern power systems—the increased load on the grid due to the rising adoption of electric vehicles. Their study revealed that uncontrolled EV charging, especially during peak hours, can significantly contribute to voltage fluctuations and frequency instability. The authors modeled various grid scenarios and found that localized clustering of EV chargers created hotspots of excessive demand. They emphasized the need for demand-side management and smart grid technologies to mitigate this issue. Without proper load scheduling, grids in urban areas may face transformer failures and increased line losses. Additionally, the quality of power delivered to other consumers may deteriorate due to harmonics introduced by uncoordinated charging systems. The research also suggested time-of-use pricing as a possible control strategy

Singh and Saini [7] – Infrastructure Costs in Rural Expansion

Singh and Saini focused on the economic and logistical challenges of expanding grid infrastructure to meet the growing demand for EV charging in rural and remote regions. Their study emphasized that installing new distribution lines, transformers, and substations entails a massive capital investment. In low-density population areas, the return on investment is often

too low to justify such expenditures. They proposed decentralized renewable energy systems, such as solar-powered chargers, as cost-effective alternatives. Furthermore, they identified bureaucratic delays and land acquisition issues as significant hurdles in grid expansion. The paper also compared the lifecycle costs of grid-based versus solar-based systems, with the latter proving more feasible in long-term deployment. Singh and Saini also pointed out the lack of skilled personnel in rural areas, which hampers the maintenance and operation of complex grid-based infrastructure. Their recommendations supported hybrid systems that combine grid and renewable sources. This approach could bridge the gap while minimizing capital outlays.

Jain et al. [8] – Grid Intermittency and Charging Reliability

Jain et al. studied the issue of intermittent and unreliable grid power, particularly in developing countries. They found that frequent blackouts and voltage sags directly affect the charging process and can cause battery damage due to inconsistent input. Their research included a statistical analysis of power outage durations in several regions and correlated them with user complaints regarding EV charging. To overcome this challenge, the authors proposed a hybrid system where solar energy is stored in local battery units and used to supplement grid power. Their simulation results showed that this method provided a more consistent voltage profile to the EV charger. They also emphasized the need for intelligent energy management systems that automatically switch between grid and solar sources based on availability. Jain et al. concluded that reliability is a major concern in grid-dependent charging and must be addressed to boost user confidence. Their study supports the integration of renewables for both energy resilience and battery protection.

Prasad and Rao [9] – Carbon Emissions and Fossil Fuel Dependency

Prasad and Rao explored the paradox that, while EVs are marketed as clean alternatives, grid-based charging systems may still rely heavily on fossil fuel-powered electricity. Their life-cycle assessment revealed that EVs charged from coal-based grids could still contribute significantly to greenhouse gas emissions. The authors pointed out that merely shifting emissions from tailpipes to power plants does not solve the core issue of environmental sustainability. They also discussed how the carbon intensity of electricity generation varies by region, making grid-based EVs less green in some areas compared to others. The study recommended that EV infrastructure development be coupled with renewable energy expansion. The authors further proposed policy-level changes, such as carbon taxes and subsidies for green charging stations. Their conclusions underscored the importance of source transparency—knowing where the electricity is coming from is as important as how it is used. This paper plays a pivotal role in evaluating the true environmental cost of EV adoption.

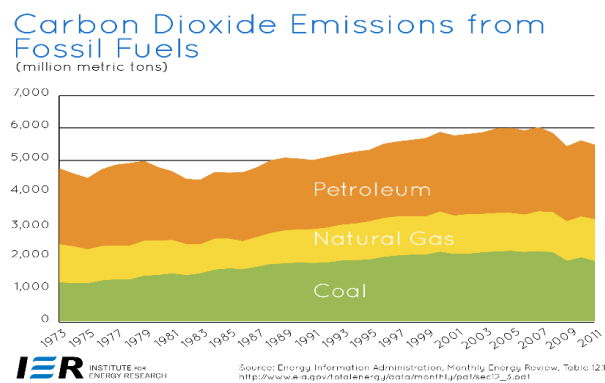


Fig.2.1 Carbon Dioxide Emission

2.3 Additional Relevant Studies

Several studies have explored the development and deployment of solar-powered EV charging systems and the associated technical, economic, and environmental challenges. This section presents selected works that are directly relevant to solar integration, energy flow control, and battery safety in electric vehicles. These insights contribute to the foundational basis of the proposed 48V solar-based e-bike charger with IoT protection.

2.3.1 Ghosh et al. – Solar-Based Charging System for EVs

In their work, Ghosh et al. (2021) analyzed various photovoltaic (PV) system configurations suitable for EV charging applications. They proposed a grid-independent solar charging system that combines a solar panel array, charge controller, battery storage, and power conversion units. The paper emphasized the need for optimizing the size of solar panels based on solar irradiance profiles and vehicle load requirements. The relevance to the current project lies in the discussion of fluctuating solar input and the need for efficient energy storage, which aligns with the challenges faced while designing a standalone solar e-bike charger.

2.3.2 Yousuf et al. – Impact of Grid Load and Solar Integration

Yousuf et al. (2020) conducted a comparative study of grid-only EV charging and solar-assisted EV charging systems. They highlighted that grid-based EV charging contributes to increased peak demand, voltage instability, and grid stress, especially in densely populated areas with poor power infrastructure. The study recommended partial or full solar integration as a way to reduce dependency on grid electricity and minimize energy losses due to transmission and distribution. Although the study also considered grid-connected hybrid systems, its discussion on the drawbacks of grid dependency underlines the importance of standalone solar solutions in areas with frequent outages or limited infrastructure—precisely the conditions targeted by this project.

2.3.3 Patel and Kulkarni – Design and Cost Evaluation of Solar Chargers

In another relevant study, Patel and Kulkarni (2019) explored the practical design of a low-cost solar charging station for two-wheelers. Their system utilized a regulated DC-DC converter and simplified control circuitry to minimize cost while maintaining acceptable charging performance. The authors provided a detailed cost-benefit analysis, highlighting the viability of small-scale solar charging systems for rural and semi-urban deployment. The study supports the design decisions made in this project by advocating for cost-effective, scalable, and easily maintainable solar charging stations, especially for low-voltage vehicles such as e-bike

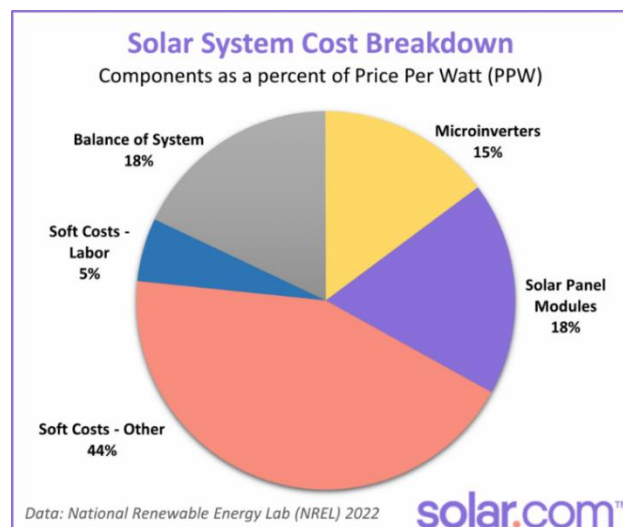


Fig.2.2 Solar System Cost Breakdown

2.4 Observations from Literature and Scope for Improvement

2.4.1 Key Insights from Literature

- **Solar Integration is Feasible and Beneficial:** Multiple studies confirm that solar energy is a viable alternative to grid electricity for EV charging, especially in sun-rich regions. It offers environmental benefits and alleviates grid stress.
- **Grid Charging Faces Infrastructure and Reliability Issues:** Research highlights the strain that large-scale EV charging puts on power grids, including overload risks and voltage instability. These problems make solar-based off-grid systems more attractive in regions with unreliable power supply.
- **Battery Protection is Often Overlooked in Solar Systems:** While many works focus on energy generation and power electronics, fewer emphasize integrated battery monitoring and protection—especially for lower-voltage systems like e-bikes.

- Complexity vs. Practicality Trade-off: Studies that use sophisticated controllers and algorithms (e.g., MPPT) achieve better performance but at higher costs and complexity. This trade-off is especially critical in low-budget and small-scale projects.

2.4.2 Scope for Improvement in Existing Systems

Despite progress in the field, certain limitations remain unaddressed:

- Lack of Smart Monitoring in Low-Cost Systems: Most small-scale solar chargers lack features like real-time battery health monitoring or user alerts. Integrating IoT-based protection can significantly enhance safety and usability.
- Overemphasis on High-Voltage EVs: A significant portion of existing research focuses on electric cars, while fewer studies target solutions for 48V systems used in personal e-bikes or scooters. Our project helps fill this gap.
- Simplified but Safe Charging Systems are Needed: There is room for systems that forgo complex algorithms but maintain user safety and energy reliability through practical, hardware-based implementations.

2.4.3 How This Project Addresses the Gaps

This project proposes a novel yet simple 48V solar charger for e-bikes, equipped with IoT-based protection features. The design includes:

- Dedicated Sensor Monitoring: Continuous real-time sensing of temperature and voltage to detect abnormal conditions. IoT Alert System: Users receive notifications through cloud-based platforms if unsafe charging conditions occur.
- Standalone Solar Operation: Completely avoids grid dependency, enhancing reliability in off-grid locations. Cost-Effective, Scalable Design: Uses basic components and protection circuitry to maintain affordability without compromising safety.

By addressing these specific gaps, the project positions itself as a practical solution for safe, solar-based charging in low-voltage electric mobility.

CHAPTER 3

CHAPTER 3

System Requirements

3.1 Introduction

The design and execution of a reliable embedded system for electric vehicle (EV) charging demand the careful selection of suitable hardware components and supportive software platforms. In the case of a solar-powered 48V e-bike charger with IoT-based battery protection, the integration of various technologies is crucial to achieve intelligent control, real-time monitoring, and efficient power utilization. System requirements not only define the capabilities of the project but also influence its scalability, performance, and operational safety.

This chapter presents a detailed overview of both hardware and software resources essential for the implementation of the proposed system. The hardware section encompasses components such as sensors, microcontrollers, power supplies, and communication modules, all of which contribute to data acquisition, decision-making, and safe power delivery. The software tools, on the other hand, enable programming, data transmission, and cloud-based monitoring.

By outlining these requirements, this chapter sets the foundation for understanding how each element contributes to the system's functionality. Proper component selection ensures that the system meets its intended goals of providing a cost-effective, sustainable, and intelligent EV charging solution. The following sections categorize the requirements into two major parts — software and hardware — based on their functionality and role in the project.

3.2 Hardware and Software Requirements

3.2.1 Software Requirements

- Embedded C Programming Language
- Arduino IDE Compiler
- Blynk Cloud Server

3.2.2 Hardware Requirements

- LCD Module
- Arduino Controller
- GSM Modem
- NodeMCU
- 48V Battery

- Solar Panel
- 12V Battery
- 12V / 7Ah Battery
- Voltage Regulators
- Filter Capacitors
- Voltage Sensor
- Temperature Sensor
- Other Miscellaneous Components (e.g., resistors, transistors, relays, boost converter, diodes)

3.3 Software Description

3.3.1 Arduino IDE and Software Implementation

To program and control the microcontroller units used in the project (Arduino and NodeMCU), the **Arduino Integrated Development Environment (IDE)** is employed. The Arduino IDE is a free, open-source platform that simplifies the process of writing, compiling, and uploading code to Arduino-compatible boards. It is available for Windows, Mac, and Linux operating systems and can be downloaded directly from the official website <https://www.arduino.cc/en/software>. After downloading, the installation process is simple and includes USB drivers necessary for communication with the Arduino or NodeMCU boards. Once installed, the IDE allows users to write embedded C programs, select appropriate boards from the Tools menu, and upload compiled sketches to the hardware through a USB interface. The IDE also comes with a rich collection of example programs that help beginners understand how to interact with sensors, displays, communication modules, and other peripherals. This makes it a highly recommended environment for embedded development.

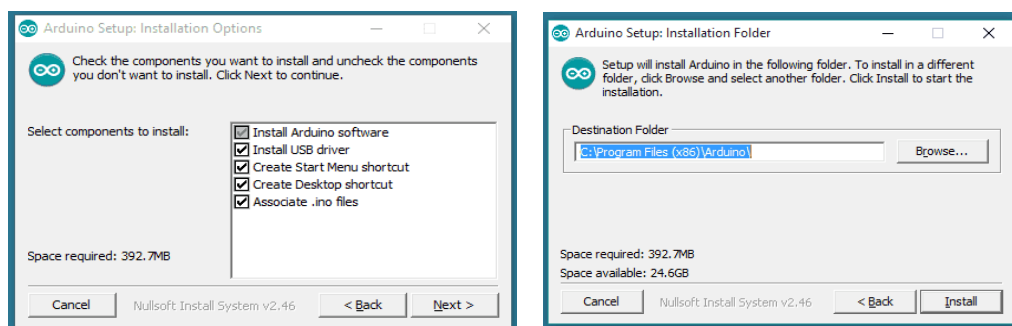


Fig.3.1&3.2-Installation guide of Arduino UNO

3.3.2 Embedded Systems Overview

An embedded system is a specialized computing system that performs a dedicated function within a larger mechanical or electrical system. Unlike general-purpose computers, embedded systems are optimized for specific tasks, often with real-time computing constraints. These systems are found in everyday devices such as washing machines, smart thermostats, automotive control units, and industrial machinery.

As illustrated in the figure below, the process begins with a real-world problem, which is first addressed through human instructions written in a high-level programming language. These instructions are then compiled into machine-level binary code that a microcontroller can execute. Finally, the system processes the instructions and performs the required actions to solve the problem.

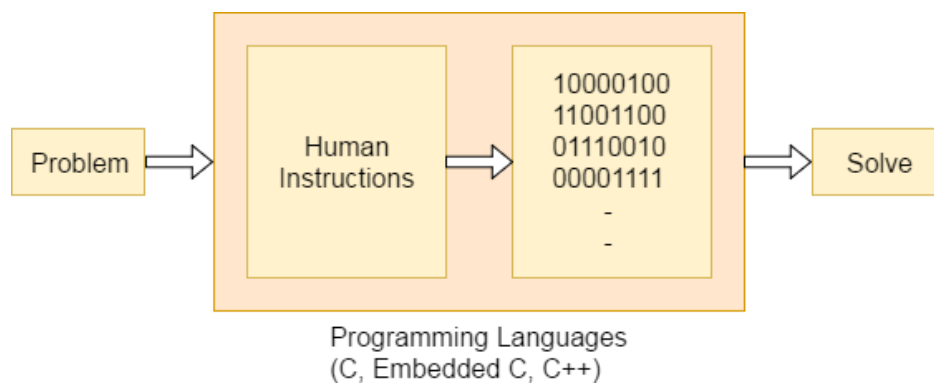


Fig.3.3 Programming Language working

This diagram simplifies how embedded systems convert logical instructions into executable code, forming the bridge between user intent and automated action. In essence, it demonstrates the basic operational workflow of an embedded system.

3.3.3 Embedded C Programming Language

Embedded C is an extension of the standard C programming language tailored for developing applications on embedded hardware. It supports features such as hardware-level I/O access, fixed-point arithmetic, and optimized memory management—making it ideal for resource-constrained environments like microcontrollers.

As illustrated in the figure below, the development process in Embedded C typically starts with identifying the **problem**, breaking it down into **step-by-step procedures**, and using **flowcharts or logic diagrams** to visualize control flow. These structured instructions are then converted into **human-readable code** and finally translated into **machine-level binary instructions** that the microcontroller can understand and execute.

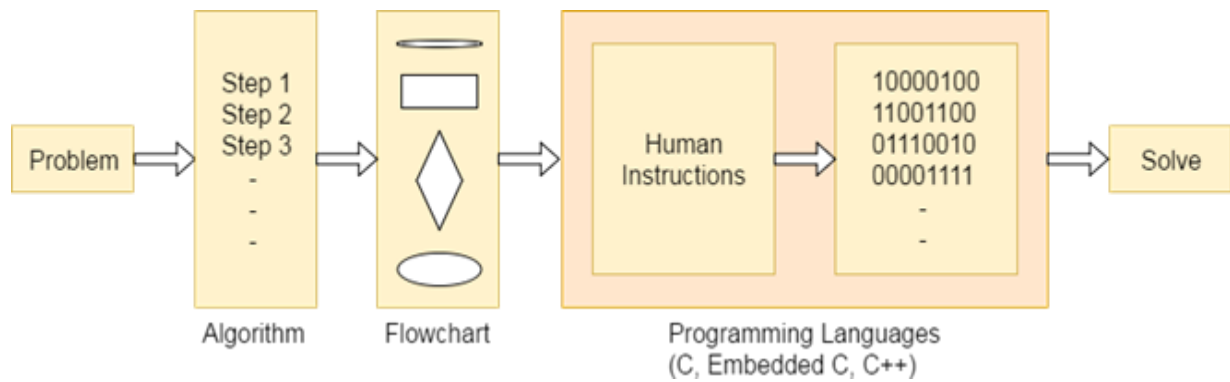


Fig.3.4 Embedded C workflow

This diagram clearly shows how problem-solving in embedded systems evolves from conceptual planning to low-level execution, emphasizing the importance of Embedded C in this pipeline. Embedded C is widely adopted due to its simplicity, portability, and close-to-hardware syntax. The structure of Embedded C programming includes standard elements like:

- `main()` function
- Conditional statements (if, switch)
- Loops (for, while)
- Bit manipulation
- Header files
- Input/output functions

In this project, the microcontroller is programmed using Embedded C through the Arduino IDE. This allows the system to monitor real-time parameters such as battery voltage and temperature and perform logical operations to control the charging mechanism and safety alerts effectively.

3.4 Hardware Description

The hardware components used in this project form the backbone of the solar-based 48V e-bike charger with IoT-based protection. Each component plays a vital role in achieving reliable energy harvesting, efficient power conversion, real-time monitoring, and user interfacing. From microcontrollers like Arduino UNO and NodeMCU to sensors and regulators, every module contributes to functionality, safety, and communication.

This section outlines and explains all the major hardware elements used, starting from the control units and display modules to sensing devices and power regulation units. Visuals and

tables are included to support the textual content and provide clarity in understanding connections and specifications.

3.4.1 Arduino UNO

The **Arduino UNO** is a microcontroller development board built around the **ATmega328P** microcontroller chip. It is specifically designed to simplify the process of embedding electronics into multidisciplinary projects. Its simple architecture and open-source design make it ideal for both prototyping and final implementations. The board includes **14 digital input/output (I/O) pins**, **6 analog input pins**, a **16 MHz crystal oscillator**, a **USB port** for programming and communication, a **power jack**, and an **ICSP header** for direct microcontroller programming. It also features an onboard **reset button** for reinitializing the system when required.



Fig.3.5 Arduino UNO Board

The board can be powered either through a **USB connection** or via an **external power supply**, such as a battery or AC-to-DC adapter. The recommended input voltage range is **7V to 12V**, but the board can safely operate between **6V and 20V** due to its onboard voltage regulator. Internally, the board runs at **5V**, which is standard for most sensors and modules. Communication protocols supported include **UART (serial communication)**, **SPI (Serial Peripheral Interface)**, and **I2C (Inter-Integrated Circuit)**, allowing the Arduino UNO to interface with a wide variety of sensors, displays, communication modules, and actuators. Due to its **open-source nature**, the Arduino UNO is extensively used in academic projects, hobbyist applications, and even industrial automation. It is compatible with the **Arduino IDE**, a beginner-friendly development environment that allows programming in **C and C++**. The board's firmware can be easily updated, and countless community-shared libraries make peripheral interfacing simple, from LCDs to temperature sensors, motor drivers, and more.

In this project, the Arduino UNO acts as the **central processing unit**, receiving data from voltage and temperature sensors, processing the information in real-time, and executing decisions such as switching relays ON/OFF, displaying output on the LCD screen, and triggering alerts via buzzer or GSM module. Its flexibility, robustness, and ease of use make it an ideal choice for this solar-powered 48V e-bike charger with IoT-based protection.

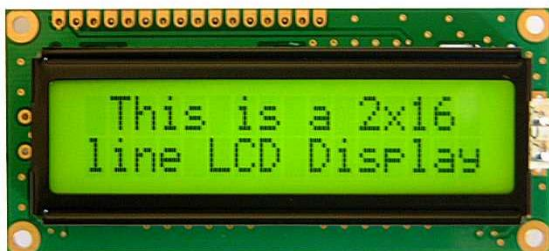
Table : Arduino UNO Pin Description

Pin Name	Function	Description
Digital 0–13	Digital I/O Pins	Can be used for input/output
PWM Pins	D3, D5, D6, D9, D10, D11	Used for analog-like output (PWM)
A0–A5	Analog Input Pins	For reading analog sensors
GND	Ground	Common ground
5V / 3.3V	Power Output	Power supply for modules
Vin	Input Voltage	External power input
TX / RX	Serial Communication Pins	For UART communication
RESET	Reset Pin	Used to reset the board

Fig.3.6 Arduino UNO Pin Description

3.4.2 LCD Display

The **LCD 16x2 module** is one of the most widely used **display components** in embedded systems and electronics projects. The term “16x2” indicates that the display consists of **16**



characters per line and **2 lines**, allowing a total of 32 characters to be displayed simultaneously. Each character is shown using a **5x7 pixel matrix**, and the display supports both **alphanumeric characters** and **custom symbols**, which makes it

suitable for a wide range of applications from digital clocks to sensor readouts and status indicators.

The module operates on **liquid crystal display (LCD) technology**, which manipulates liquid crystals using electrical signals to control the passage of light. This makes LCDs **energy-efficient** and well-suited for **battery-powered** and **portable applications**. The 16x2 LCD is a **monochrome display**, often equipped with a **backlight** (usually white, blue, or green) to enhance visibility in low-light environments. LCD 16x2 modules use a **parallel interface** and have **16 pins** in total. The most important control pins are **RS (Register Select)**, **RW**

(Read/Write), and **E (Enable)**. RS distinguishes between command and data operations, RW selects between reading from or writing to the LCD, and E is used to latch in the data. There are 8 data lines (**DB0–DB7**), although in most applications, the module is operated in **4-bit mode** (using only DB4–DB7) to reduce the number of microcontroller pins required. The sample LCD display as shown in **fig.3.7**. The remaining pins include **Vcc** and **GND** for power, **VEE** for contrast adjustment (usually connected via a potentiometer), and **LED+ / LED-** for controlling the backlight. The module typically operates at **5V** and is compatible with most microcontrollers, including Arduino, PIC, and ARM-based systems. The LCD 16x2 has **two internal registers**: the **command register** (used to configure display settings) and the **data register** (used to send characters for display). Commands such as clearing the screen, setting the cursor, and turning the display on/off are sent to the command register. ASCII values of characters are sent to the data register to display them on the screen. With **simple interfacing**, **low power consumption**, and **wide compatibility**, the LCD 16x2 remains a core component for visual output in embedded applications. Its ease of programming using libraries like LiquidCrystal.h in Arduino IDE makes it accessible even to beginners.

Table: 16x2 LCD Pin Description

Pin	Name	Function
1	VSS	Ground
2	VDD	+5V Power Supply
3	VO	Contrast Adjustment
4	RS	Register Select
5	RW	Read/Write
6	E	Enable
7–14	D0–D7	Data Pins
15	LED+	Backlight Anode (usually 5V)
16	LED-	Backlight Cathode (Ground)

Fig.3.8 LCD Pin Description

3.5 Relay

3.5.1 Overview

A relay is an electrically operated switch that allows control of a high-voltage or high-current circuit by a low-power electrical signal. It is widely used in automation and protection systems where isolation between the control and load circuit is necessary. In this project, a relay is implemented to control the charging process of a 48V e-bike battery. The IoT-based monitoring system collects real-time data such as battery voltage and temperature, and based on predefined safety thresholds, the Arduino decides whether to activate or deactivate the charging circuit via

the relay. This ensures that charging only occurs under safe operating conditions, preventing risks such as overheating, overvoltage, or shortcircuits.

3.5.2 Working Principle

The fundamental working principle of an electromagnetic relay relies on the generation of a magnetic field when electric current flows through a coil. This magnetic field attracts a movable armature, which in turn changes the state of electrical contacts. These contacts can be Normally Open (NO), Normally Closed (NC), or a combination, depending on the relay type. When the coil is energized, the magnetic field pulls the armature, either closing the NO contacts or opening the NC contacts. Once the coil is de-energized, the armature returns to its original position via a spring mechanism, restoring the contact to its initial state. In the context of this project, when the Arduino detects unsafe charging conditions—such as high temperature or voltage levels—it sends a signal to deactivate the relay, cutting off power to the battery. Conversely, under safe conditions, the Arduino energizes the coil, allowing the charging current to flow. This reliable switching mechanism ensures that control logic is effectively translated into physical circuit operation. The sample relay diagram shown in **fig.3.9**.

3.5.3 Types and Selection

There are several types of relays including SPST (Single Pole Single Throw), SPDT (Single Pole Double Throw), DPDT (Double Pole Double Throw), and solid-state relays. For this



project, an **SPDT** (Single Pole Double Throw) relay is selected. This type of relay has one common terminal and two outputs, allowing the circuit to switch between two different paths. The SPDT relay was chosen for its flexibility and ability to reliably toggle the battery charging path based on the control logic provided by the Arduino. The selection criteria also included coil voltage compatibility with Arduino (5V), current

handling capability (at least 10A for safety margin), and physical size suitable for the hardware layout. Mechanical relays, such as the one used here, are more robust for handling high switching currents compared to their solid-state counterparts, which are more suited for lower current or high-speed applications. This makes the SPDT relay an ideal choice for the reliable operation of a solar-powered battery charger for an e-bike.

3.5.4 Relay Specifications

Parameter	Value
Coil Voltage	5V DC
Switching Voltage	250V AC / 30V DC
Switching Current	10A
Contact Type	SPDT
Isolation	Yes

3.6 Resistor

3.6.1 Overview

A resistor is a fundamental passive electronic component that resists the flow of electric current, thereby reducing voltage and controlling current in a circuit. It is one of the most widely used components in electronics, used in everything from simple LED circuits to complex industrial controllers. In this project, resistors serve multiple purposes such as voltage division, current limiting, and signal conditioning for microcontroller inputs. They are especially vital for protecting sensitive components like the Arduino and sensors by ensuring they operate within their specified voltage and current ratings. Further, resistors help to maintain circuit stability and reduce noise, ensuring accurate sensor readings and safe switching through the relay.

3.6.2 Working Principle

The operation of a resistor is governed by **Ohm's Law**, which states that the voltage across a



resistor (V) is directly proportional to the current flowing through it (I), with resistance (R) being the constant of proportionality: $V=I \times R$. This means that when a voltage is applied across a resistor, it limits the amount of current that can pass through. The resistor converts excess electrical energy into heat, thus controlling the power in the circuit. In practical applications, resistors are used in series with LEDs to prevent them from burning out, across analog sensors to create voltage dividers, and in digital circuits to pull-up or pull-down floating pins. In this project, resistors are used to ensure that input pins on the Arduino do not float, and to condition signals from temperature and voltage sensors. These precise roles help

maintain safe and consistent operation across all modules of the charger system. The sample resistor diagram as shown in **fig.3.10**.

3.6.3 Types and Selection

Resistors come in various types including **carbon film, metal film, wire-wound, and surface mount (SMD)**. For this project, **carbon film and metal film resistors** are primarily used due to their reliability, and cost- effectiveness. Carbon film resistors are preferred for general-purpose current limiting and pull-up functions, while metal film resistors are used where precision and stability are critical, such as in voltage divider networks for sensors. Selection of resistor values was done by calculating desired current limits and voltage drops using Ohm's Law. Power rating is another key consideration; for most parts of the circuit, ¼ watt resistors were sufficient. However, in areas involving higher current, higher wattage resistors are used to prevent overheating. Tolerance levels of $\pm 1\%$ were chosen for precision circuits, ensuring the Arduino receives consistent and accurate analog values from sensors.

3.6.4 Resistor Specifications

Parameter	Value
Resistance Range	220 Ω , 1k Ω , 10k Ω
Power Rating	0.25W – 0.5W
Tolerance	$\pm 1\%$ to $\pm 5\%$
Type	Carbon Film / Metal Film
Operating Temp Range	-55°C to +155°C

3.7 Boost Converter

3.7.1 Overview

A **Boost Converter**, also known as a step-up converter, is a type of DC-DC converter that increases the input voltage to a higher output voltage while decreasing the current. It operates on the principle of energy storage in inductors and the controlled release of that energy to achieve voltage boosting. In the context of this project, the boost converter plays a critical role by converting the **12V DC** from the intermediate battery storage to **48V DC**, which is required to charge the e-bike's battery efficiently. The converter enables energy-efficient voltage transformation without the need for bulky transformers or additional power stages. It is compact,

cost-effective, and ideal for low to medium-power applications like personal electric mobility devices.

3.7.2 Working Principle

The basic working of a boost converter involves **switching ON and OFF a transistor** (usually a MOSFET) connected with an **inductor**, **diode**, and **output capacitor**. Here's a simplified explanation:

- **When the switch is ON:** Current flows through the inductor and stores energy in its magnetic field. The output load is temporarily powered by the output capacitor.
- **When the switch is OFF:** The inductor's magnetic field collapses and releases energy, which, combined with the input voltage, flows through the diode to the output capacitor and load. This increases the output voltage. This switching action is controlled by a **PWM (Pulse Width Modulation)** signal from the microcontroller or an internal controller circuit. The output voltage can be regulated by adjusting the duty cycle of the PWM.

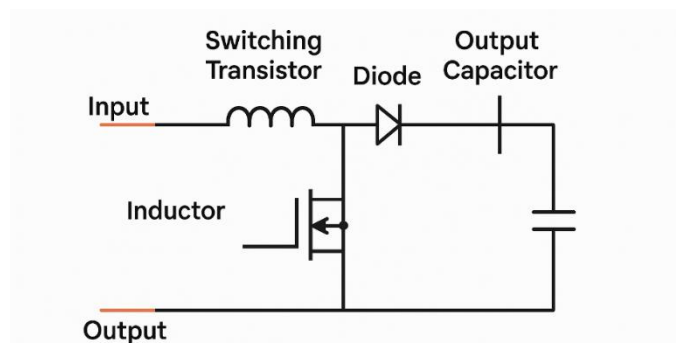


Fig.3.11 Boost Converter

3.7.3 Specifications Used in the Project

Parameter	Value
Input Voltage	12V DC (from battery)
Output Voltage	48V DC
Max Output Current	2–5 Amps (depending on module rating)
Efficiency	85% – 95%
Switching Frequency	~150 kHz (typical)
Control Method	PWM-based Regulation

These values are suitable for lightweight e-bike applications, where power requirements are moderate but voltage needs to be stepped up significantly.

3.7.4 Placement in the Circuit and Operation Flow

- The boost converter is **connected after the 12V battery and before the relay switch and charging port.**
- Once the system confirms safe voltage and temperature ranges via IoT monitoring (NodeMCU + sensors), the relay activates and allows power from the **boost converter output to reach the e-bike battery.**
- During operation, the converter must be capable of handling dynamic load changes as the battery Charge.

3.8 GSM MODEM 800

GSM stands for Global System for Mobile Communications. It is a standard set developed by the European Telecommunications Standards Institute (ETSI) to describe protocols for second generation (2G) digital cellular networks used by mobile phones. A Modem is a device which



modulates and demodulates signals as required to meet the communication requirements. It modulates an analog carrier signal to encode digital information, and also demodulates such a carrier signal to decode the transmitted information. A GSM Modem is a device that modulates and demodulates the GSM signals and in this particular case 2G signals. The modem we are using is

SIMCOM SIM800. It is a Tri-band GSM/GPRS Modem as it can detect and operate at three frequencies (EGSM 900 MHz, DCS 1800 MHz and PCS1900 MHz). Default operating frequencies are EGSM 900MHz and DCS 1800MHz. Sim800 is a widely used in many projects and hence many variants of development boards for this have been developed. These development boards are equipped with various features to make it easy to communicate with the SIM800 module. Some boards provide only TTL interface while some boards include an RS232 interface and some others include an USB interface. If your PC has a serial port(DB9) you can buy a GSM Modem that has both TTL and RS232 interfacing in economy. The sample GSM Board diagram as shown in **fig.3.12**.Sim800 GSM Module can be used to send and receive SMS connecting it to a PC when a SIM is inserted. The GSM Modem can be sent commands to send or receive SMS from the PC through a com port (serial port or an usb). These commands

are called as **AT commands**. Through AT commands we can perform several actions like sending and receiving SMS, MMS, etc. Sim800 has an RS232 interface and this can be used to communicate with the PC. Sim800 usually operates at a baudrate of 9600, with 1 stopbits, No parity, No Hardware control and 8 databits. We shall see at some of the AT Commands necessary for sending and receiving SMS.

AT Commands to Send/Receive SMS

AT+CMGF=1 => Make SIM800 to text mode

AT+CMGR=1 => To read the first message

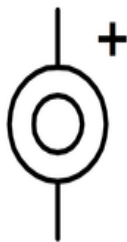
AT+CMGS="mobile number" => To send SMS

AT+CMGD=1 => To delete first message

AT+CMGD=1,4 => To delete all messages

3.9 Buzzer

A buzzer is an audio signaling device used to provide alerts or notifications through sound. It can be of mechanical, electromechanical, or piezoelectric type. Piezoelectric buzzers are the most commonly used in embedded systems due to their compact size and energy efficiency.



Buzzers are used in systems to indicate events such as power on/off, fault detection, or process completion. They work by converting electrical signals into sound using mechanical vibration or piezoelectric effect. When a voltage is applied, the piezo element deforms and generates an audible tone. The circuit Buzzer diagram as shown in **fig.3.13**.

3.10 Voltage Sensor

A **Voltage Sensor** is an essential component in embedded and monitoring systems, designed to detect and measure the electrical voltage in a circuit. These sensors are capable of reading both **AC and DC voltage levels**, and converting them into readable output signals, such as **analog voltage, current signals**, or **digital triggers** for microcontroller input. The voltage sensor module typically comprises **input and output terminals**. The input side consists of two pins: one for the **positive terminal** and another for the **negative terminal** of the voltage source to be measured. The output side generally provides:

- **Vcc (Power Supply)**
- **GND (Ground)**

- **Analog Output (A0)**

Most commonly, a **voltage divider circuit** is integrated inside the sensor to scale down the measured voltage to a microcontroller-compatible level (usually 0–5V). In this setup, resistors act as the key sensing components. The **output voltage** from the sensor can be mathematically determined by: $V_{out} = (R1 / (R1 + R2)) \times V_{in}$

This ensures the input voltage is safely reduced and transmitted to the analog pin of a controller (e.g., Arduino UNO) for processing. For more accurate and advanced versions, **bridge circuits** with four resistors may be employed to enhance detection sensitivity

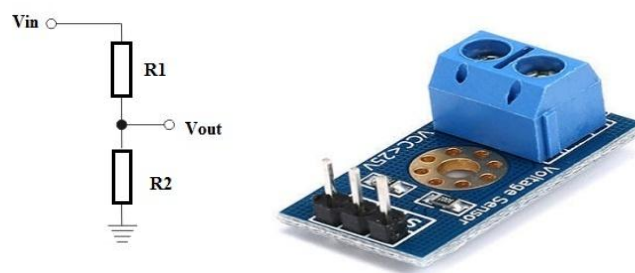


Fig.3.14 Voltage Sensor

3.11 Electrolytic Capacitor

An **electrolytic capacitor** is a polarized capacitor that utilizes an electrolyte (a conductive gel or liquid) to provide high capacitance in a compact size. These capacitors are most commonly made from aluminum or tantalum and are frequently used in circuits such as power supplies, audio filters, voltage smoothing units, and energy storage blocks. The internal construction involves two aluminum foils—one with an oxide layer acting as the anode and the other as the cathode—separated by a paper soaked in electrolyte. These components are rolled and enclosed in a cylindrical shell, forming either axial or radial packages. Because of their polarity, it is crucial that the positive terminal is always connected to a higher voltage; incorrect orientation can lead to overheating or rupture. They have relatively high ESR (Equivalent Series Resistance), making them less suitable for high-frequency applications. However, their ability to offer large capacitance in a small space makes them ideal for DC filtering and energy buffering. They are known to have wide tolerance ranges (often $\pm 20\%$) and may experience a drop in performance over time due to capacitance drift or electrolyte drying. For long storage durations, capacitor



reforming may be required to restore the dielectric layer. These components are typically rated from 1 μ F to tens of thousands of microfarads and can handle voltages up to a few hundred volts. They also feature safety vents to prevent explosion under stress. The sample capacitor diagram as shown in **fig.3.15**.

3.12 Temperature Sensor

A **temperature sensor** is an essential component in electronic systems used to detect temperature changes in the environment or in specific devices like batteries. It works by sensing the ambient or object temperature and converting it into an electrical signal that can be interpreted by a microcontroller. In embedded systems, temperature sensors like **LM35**, **DHT11**, or **NTC thermistors** are widely used due to their compact size, low cost, and high accuracy. The **LM35**, in particular, provides a linear analog voltage output directly proportional to the temperature in degrees Celsius, with a scale factor of 10mV/°C. For example, if the temperature is 30°C, the sensor will output 0.30V. This makes it easy to interface with analog-to-digital converters (ADCs) found in Arduino or NodeMCU boards.

These sensors operate on low power and are designed to be highly stable over time. In the context of EV battery systems, temperature sensors are crucial for monitoring the heat generated during the charging and discharging processes. If the temperature exceeds safe thresholds, the microcontroller can take corrective actions, such as disconnecting the power supply via a relay or triggering an alert via a buzzer or GSM message. This not only extends battery life but also prevents thermal runaway conditions, which could otherwise result in serious safety hazards. Temperature sensors are also important in real-time systems where precise thermal feedback is necessary. In this project, the temperature sensor continuously sends data to the Arduino, which processes and forwards the data to a cloud server through the NodeMCU. The real-time monitoring enables users to track battery temperature remotely and respond proactively if the temperature goes beyond the safe operating range.

These sensors can be installed in close contact with the battery casing to ensure accurate thermal readings. They are also immune to noise and external interference, especially when shielded cables and proper grounding are used. Most temperature sensors have three pins—Vcc, GND, and Signal—and are easily mountable on a PCB.

Temperature Sensor Circuit LM35

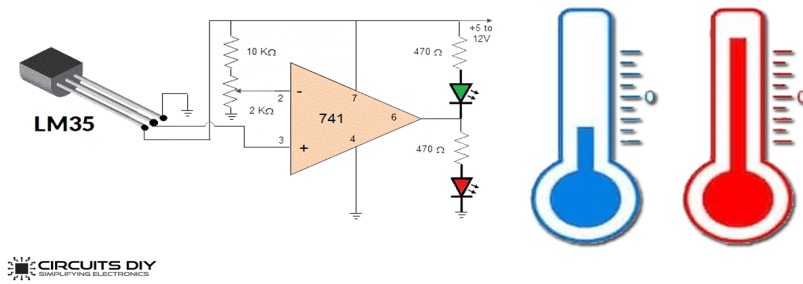


Fig.3.16 Temperature Sensor

3.13 Voltage Regulators

A **voltage regulator** is a crucial component in electronic circuits designed to maintain a consistent and stable output voltage, regardless of changes in input voltage or load conditions. These devices ensure that sensitive electronic components receive the correct voltage required for safe and efficient operation. Voltage regulators are broadly categorized into **linear** and **switching** types. In this project, linear voltage regulators from the **78xx family**—specifically the **7805** and **7812**—are employed to deliver fixed 5V and 12V outputs, respectively.

The 7805 regulator provides a **regulated 5V output**, which is essential for powering microcontrollers like **Arduino UNO**, **NodeMCU**, and sensors. The 7812, on the other hand, offers a **12V regulated output** required by modules such as relays or GSM modems. These voltage regulators have **three pins**: input voltage (V_{in}), ground (GND), and regulated output voltage (V_{out}). They are widely available in **TO-220 packages** and are appreciated for their simplicity, reliability, and low cost.

Voltage regulators help in avoiding damage to electronic components by preventing overvoltage situations, which can otherwise lead to overheating or malfunctioning of ICs and modules. The **input voltage** to a regulator should always be higher than its output voltage by a specific margin known as the **dropout voltage**, which is typically 2–3V for linear regulators. The 78xx family can handle input voltages ranging from about **7V to 35V**, and they typically supply current up to **1–1.5A**, depending on the heat dissipation and cooling mechanism used. For improved stability, **filter capacitors** are often placed at both input and output terminals of the regulator to suppress voltage fluctuations and noise.



Fig.3.17 Voltage Regulators

3.14 Solar Panel

A **solar panel**, also known as a photovoltaic (PV) module, is a crucial component in renewable energy systems. It converts sunlight directly into electrical energy through the photovoltaic effect. Each panel consists of numerous interconnected solar cells made from semiconducting materials, typically silicon. When sunlight strikes the cells, photons excite electrons, generating a flow of electric current. This direct current (DC) can then be utilized, stored in batteries, or converted into alternating current (AC) using an inverter.

Solar panels are rated based on their power output under standard test conditions (STC), usually ranging from **100 to 365 watts per panel**. The **efficiency** of a solar panel indicates how much of the incident sunlight is converted into usable electricity. Commercial panels generally have an efficiency between **15% and 22%**, although advanced laboratory panels have exceeded 40% using multijunction and concentrator technologies. The performance of a solar panel is influenced by several factors such as **solar irradiance, angle of installation, temperature, shading, and dust accumulation**. Modern solar systems use **MPPT (Maximum Power Point Tracking)** technology in inverters to continuously optimize the power output by adjusting the load and maintaining the panel's operation at its peak power point. Solar panels are available in different types, including **monocrystalline, polycrystalline, and thin-film**, each with distinct characteristics in terms of efficiency, cost, and space requirements. Monocrystalline panels are generally more efficient and space-saving, making them suitable for rooftop installations, while polycrystalline panels offer a cost-effective solution for large-scale ground-mounted systems.



Fig.3.18 Solar Panel

With the continued reduction in cost per watt and improvement in material technology, solar panels have become the cornerstone of sustainable energy solutions. They are commonly used in combination with batteries and controllers in standalone or grid-connected systems, providing a **clean, reliable, and renewable source of electricity** with minimal maintenance and long operational life.

3.15 Battery (12V / 7.2Ah and 48V)

Lead-acid batteries remain one of the most widely used rechargeable battery technologies due to their cost-effectiveness, reliability, and availability. These batteries operate through electrochemical reactions between lead dioxide (PbO_2) and sponge lead (Pb) in the presence of a sulfuric acid (H_2SO_4) electrolyte. When discharging, chemical reactions form lead sulfate (PbSO_4) on both plates, releasing electrons and generating electrical energy. During charging, this reaction is reversed, restoring the plates and electrolyte for reuse. The **12V / 7.2Ah sealed lead-acid battery** is commonly employed for moderate power storage needs. It provides a compact and maintenance-free energy source for embedded and IoT systems, emergency backup, and mobility applications. It is constructed using a valve-regulated lead-acid (VRLA) design, which prevents leakage and reduces maintenance by using an absorbed glass mat (AGM) or gel separator to immobilize the electrolyte. This battery delivers a nominal current of **350 mA at a 20-hour discharge rate**, making it suitable for energy buffering in solar-based systems. The sample battery diagram as shown in fig.3.1. The **48V battery** is a higher-voltage configuration achieved by connecting multiple 12V batteries in series. This voltage level is ideal for electric vehicles like e-bikes, where higher energy density and sustained current output are required to drive motor loads efficiently. The higher voltage enables reduced current for the same power, thus minimizing conductor size and system losses. Like its 12V counterpart, the 48V system typically uses sealed lead-acid cells for safety and durability. Key features of these

batteries include **sealed and maintenance-free operation**, **deep discharge recovery**, and **low self-discharge rates**, allowing them to be stored for long periods without significant energy



loss. They also offer a **long cycle life**, supporting **200–1000 charge/discharge cycles**, depending on depth of discharge and operating conditions. Their **high discharge rate** makes them capable of handling peak loads several times their nominal rating, a critical attribute for starting motors and handling transient loads. These batteries are suitable for use in

a wide **operating temperature range from -60°C to +60°C**, enabling deployment in various environmental conditions.

3.16 NODE MCU

NodeMCU Dev Kit/board consists of ESP8266 WiFi enabled chip. The ESP8266 is a low-cost



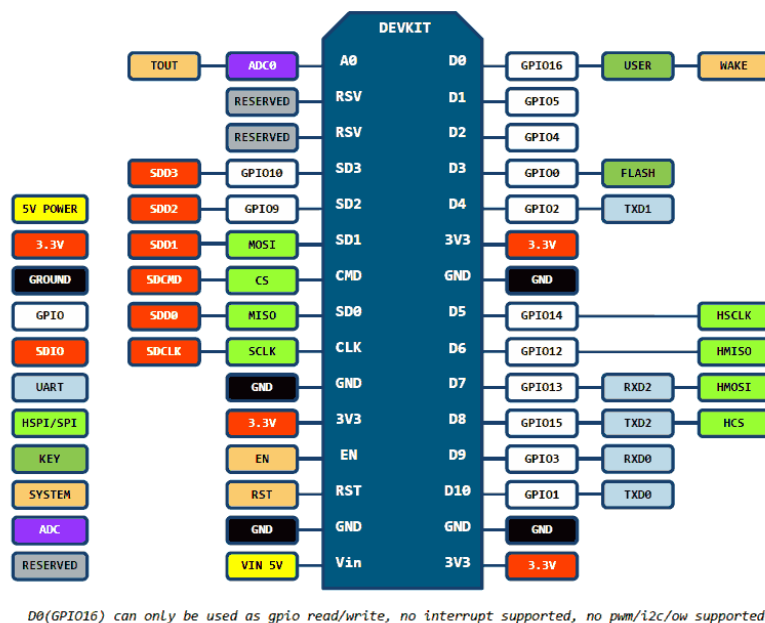
Wi-Fi chip developed by Espressif Systems with TCP/IP protocol. For more information about ESP8266, you can refer ESP8266 WiFi Module. There is Version2 (V2) available for NodeMCU Dev Kit i.e. NodeMCU Development Board v1.0 (Version2), which usually comes in black colored PCB.

NodeMCU Dev Kit has Arduino like Analog (i.e. A0) and Digital (D0-D8) pins on its board. It supports serial communication protocols i.e. UART, SPI, I2C etc. Using such serial protocols we can connect it with serial devices like I2C enabled LCD display, Magnetometer HMC5883, MPU-6050 Gyro meter + Accelerometer, RTC chips, GPS modules, touch screen displays, SD cards etc. NodeMCU Development board is featured with WiFi capability, analog pin, digital pins and serial communication protocols. To get started with using NodeMCU for IoT applications, first we need to know about how to write/download NodeMCU firmware in NodeMCU Development Boards. And before that, where this NodeMCU firmware will get as per our requirement. There is online NodeMCU custom builds available using which we can easily get our custom NodeMCU firmware as per our requirement. The Sample NODE MCU chip as Shown in **Fig.3.20**.

Development Board

NodeMCU Development Kit/Board consist of ESP8266 wifi chip. ESP8266 chip has GPIO pins, serial communication protocol, etc. features on it. **ESP8266** is a low-cost [Wi-Fi](#) chip developed by Espressif Systems with TCP/IP protocol. For more information about ESP8266, you can refer [ESP8266 WiFi Module](#). The features of ESP8266 are extracted on NodeMCU Development board. NodeMCU ([Lua](#) based firmware) with Development board/kit that consist of ESP8266 (wifi enabled chip) chip combines NodeMCU Development board which make it stand-alone device in IoT application.

PIN DEFINITION



CHAPTER 4

CHAPTER 4

System Design

4.1 Introduction

The system design phase is a crucial step in the development of any hardware-based engineering project, as it lays the foundation for the successful integration and operation of all components. In this project, a solar-based 48V e-bike charger with IoT-enabled protection is developed to promote the use of clean energy and enhance the safety and efficiency of EV charging. The design involves a well-organized arrangement of key components such as solar panels, charge controllers, batteries, sensors, microcontrollers, and IoT modules, each playing a vital role in the overall operation.

Proper system design ensures that energy is harvested efficiently from the sun and stored safely in the battery bank before being used to charge the e-bike. Sensors continuously monitor critical parameters like voltage and temperature, which are then processed by the microcontroller to ensure the safe operation of the system. The IoT module provides wireless data transmission, enabling remote tracking and analysis of system performance. Additionally, protection mechanisms are embedded into the design to prevent overcharging, overheating, or system failures. The design also considers real-world challenges such as fluctuating solar input, battery health management, and environmental conditions. A systematic approach is used to reduce energy losses, improve component lifespan, and maintain stability across the entire system. Overall, this chapter describes how each component is selected and positioned logically within the system to achieve maximum reliability, performance, and user convenience.

4.2 Block Diagram of the System

The block diagram provides a simplified visual representation of the complete system architecture. It outlines the flow of power and data between key components, beginning from the solar panel to the final e-bike charging interface. Each block denotes a distinct functional unit, including the power supply unit, Arduino controller, relay, sensors, GSM and Wi-Fi modules, buzzer, LCD, and motor.

The connections between the blocks show how data and energy move through the system in a controlled and monitored manner. This modular representation helps in understanding how

each component interacts with the others. The diagram also highlights safety and control elements, such as manual switches, communication interfaces, and feedback loops essential for real-time monitoring and system protection.

System Architecture (Block Diagram)

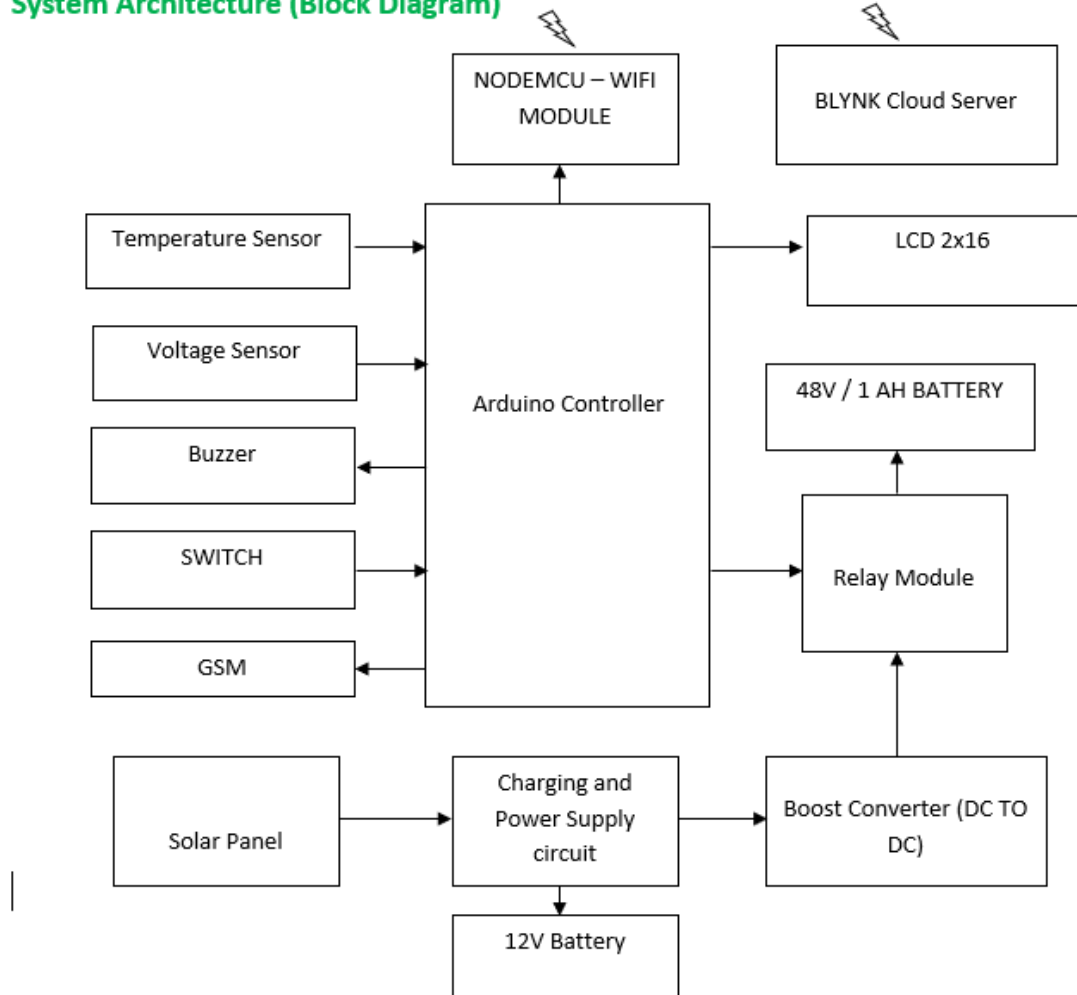


Fig.4.1 Block Diagram

- **Solar Panel:** This is the primary energy source, harvesting solar energy and supplying it to the charging and power supply circuit.
- **Charging and Power Supply Circuit:** This unit regulates the incoming power and stores it in a 12V battery, acting as a buffer energy source.
- **12V Battery:** Stores the regulated power from the solar panel. The energy from this battery is later boosted to the required 48V using a DC-DC converter.
- **Boost Converter (DC to DC):** Converts 12V DC to 48V DC, making it suitable for charging the 48V e-bike battery.

- **Relay Module:** Acts as an electronic switch, controlled by the Arduino, to enable or disable charging to the 48V battery for safety and automation purposes.
- **48V / 1Ah Battery:** This is the e-bike battery which gets charged from the boosted 48V power supply.
- **Arduino Controller:** The central processing unit that collects data from various sensors, makes decisions, and controls the relay, buzzer, display, and communication modules.
- **Temperature Sensor:** Monitors the temperature of the battery or the circuit. If abnormal heat is detected, it triggers safety mechanisms.
- **Voltage Sensor:** Continuously monitors the voltage levels during charging and discharging to prevent overvoltage or undervoltage conditions.
- **Buzzer:** An alert system that activates during faults such as overcharging or overheating.
- **Switch:** Provides manual control over system operations when needed.
- **GSM Module:** Used for mobile communication or alerts via SMS (if implemented).
- **NodeMCU – Wi-Fi Module:** Enables wireless connectivity, sending real-time data to the Blynk Cloud Server.
- **Blynk Cloud Server:** A cloud-based IoT platform used for real-time monitoring of system parameters remotely via a smartphone app.
- **LCD 2x16 Display:** Displays local status updates such as voltage, temperature, and charging status for user convenience.

This block diagram emphasizes a well-integrated design that ensures safety, reliability, and smart monitoring of the entire charging system using renewable energy and IoT technology.

4.3 Circuit Diagram

The circuit diagram for the Solar-Based 48V E-Bike Charger with Protection Using IoT illustrates the complete working setup of energy generation, storage, conversion, and intelligent control. It integrates solar energy harvesting, voltage regulation, battery storage, and smart monitoring using microcontrollers. The diagram, as shown in Figure 3.1, is divided into multiple functional blocks that work together to ensure safe and efficient charging of a 48V e-bike battery. It includes a solar panel, voltage regulators, a 12V battery, a boost converter, an Arduino-based control system, and protection elements like sensors, a GSM module, a buzzer, and a relay mechanism. Each block has been carefully designed and interconnected to maintain the safety, efficiency, and smartness of the system. The use of microcontrollers also enables

remote monitoring and alerts in case of faults. Overall, this circuit ensures both reliable energy delivery and protective features to prevent battery damage.

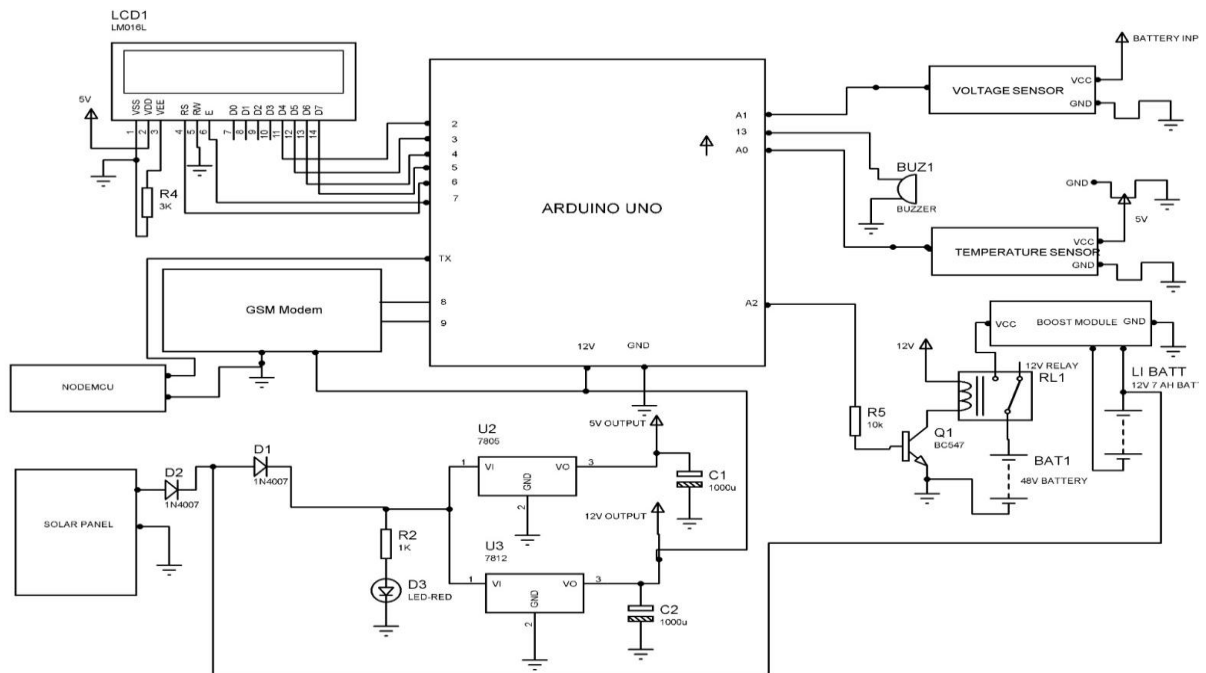


Fig.4.2 Circuit Diagram

Solar Panel and Diode Protection

The circuit begins with a solar panel, which serves as the main energy source by converting solar radiation into electrical energy. The current generated flows through two protective diodes (D1 and D2 – both 1N4007). D2 prevents the reverse flow of current from the battery back to the solar panel, especially during nighttime or low-light conditions, protecting the panel. D1 ensures unidirectional current flow towards the battery, avoiding battery discharge when the panel voltage is low. These diodes enhance reliability by securing the system from reverse currents. This setup is critical in solar charging applications where sunlight is intermittent. It also ensures the energy generated is stored efficiently without losses. The arrangement helps in maintaining the longevity of both the solar panel and battery.

Voltage Regulation and Filtering

The raw voltage from the solar panel is regulated using two linear voltage regulators: 7812 and 7805. The 7812 IC provides a stable 12V output required to power the relay, the boost converter, and other modules. The 7805 IC supplies a steady 5V to power logic-level devices like the Arduino UNO, NodeMCU, sensors, and the LCD display. Two capacitors (C1 and C2, each 1000 μ F) are connected to smooth out voltage fluctuations and eliminate ripples, ensuring clean DC power. These regulators play a crucial role in delivering stable voltage despite variations in solar output. They safeguard sensitive components from voltage spikes or drops. The inclusion

of filtering capacitors ensures system stability and performance. Overall, this regulation block is vital for maintaining consistent system behavior.

Battery Storage

The energy regulated from the solar panel is used to charge a 12V lead-acid battery. This battery functions as an energy reservoir and ensures a stable power supply to the boost converter. Even if solar input fluctuates due to weather conditions, the battery ensures uninterrupted power availability for the e-bike charging process. It acts as an intermediary storage that enables better energy management. This design ensures that high-voltage charging is not directly dependent on real-time solar output. The battery also prevents sudden shutdowns of the system due to momentary power drops. Using a lead-acid battery is cost-effective and provides sufficient capacity for this application. Moreover, it helps in maintaining voltage levels during load changes.

Boost Converter Module

A boost converter is used to step up the 12V battery output to 48V, which is required to charge the e-bike battery. It ensures that the output voltage is regulated and maintained even when the input voltage from the battery varies. The boost converter enhances the system's flexibility by allowing charging from a lower voltage source. It ensures efficient power transfer with minimal losses. This step-up mechanism is crucial since direct charging from the solar panel would not provide the necessary voltage. The boost converter also includes control circuitry to regulate the current flow. It makes the overall design compact and suitable for electric vehicle applications. This module forms the core of high-voltage energy delivery in the system

Relay and Switching Circuit

To control when the 48V charging output is delivered to the e-bike, a relay-based switching circuit is used. A 12V electromagnetic relay is connected to a BC547 NPN transistor, which acts as an electronic switch. The base of the transistor is connected through a 10K Ω resistor to limit the current and protect the transistor. The Arduino UNO sends a signal to the transistor, turning the relay ON or OFF based on sensor inputs. This setup ensures that charging only occurs under safe voltage and temperature conditions. The relay isolates the high-power charging line from the control logic. It also provides a secure means of switching without manual intervention. This smart switching enhances safety and automation.

Arduino UNO Controller

The Arduino UNO acts as the central processing unit for the system, monitoring all sensor data and executing control decisions. It reads analog signals from a voltage sensor and a temperature

sensor to continuously evaluate the battery's status. Based on the sensor readings, it determines whether to activate the relay for charging or disconnect the battery in case of faults. The Arduino also sends data to the LCD for real-time display and to the NodeMCU for cloud updates. It plays a critical role in decision-making, protection, and automation. With multiple I/O pins, the Arduino efficiently handles various sensors, output devices, and communication modules. Its programming can be customized for threshold limits and alert logic. Thus, it enables a highly responsive and intelligent system.

LCD Display

The system includes a 16x2 LCD (LM016L) to provide real-time feedback to the user. It is connected to the Arduino UNO through digital pins 2 to 7. The LCD displays current battery voltage, temperature, and charging status, making it easy to monitor the system without connecting to external devices. It helps users understand whether the system is charging, idle, or encountering a fault. This enhances user interaction and trust in the system. The LCD updates dynamically as new sensor readings are processed. It is especially useful for quick checks during testing or operation. Overall, it offers a simple yet effective interface for real-time system monitoring.

GSM and IoT Monitoring

A GSM module is used to send SMS alerts when abnormal conditions like overvoltage or overheating are detected. It is connected to the Arduino via digital pins 8 (TX) and 9 (RX). In addition, a NodeMCU module enables IoT functionality by sending sensor data to an online dashboard. This allows remote users to track the system's performance in real time through a mobile app or web browser. Alerts can be generated for predefined thresholds, improving safety. This dual communication system—GSM for alerts and NodeMCU for monitoring—ensures comprehensive awareness of system status. It adds a layer of intelligence and remote supervision to the e-bike charging setup. This makes the system truly smart and user-friendly.

Buzzer Alert System

For immediate local alerts, a piezo buzzer is connected to digital pin 13 of the Arduino UNO. When the system detects unsafe voltage or temperature levels, the Arduino triggers the buzzer to emit an audible sound. This notifies nearby users about potential hazards even if they aren't watching the display. It provides a quick and effective warning method without requiring external connectivity. The buzzer complements the GSM and IoT alerts, covering both remote and onsite users. It is a cost-effective way to ensure prompt user attention. The inclusion of a

buzzer increases the system's safety profile and responsiveness. It helps avoid damage by enabling faster human intervention.

4.4 Working Principle of the System

The proposed system introduces an innovative approach for charging an electric vehicle (EV) using solar energy and real-time monitoring through IoT technology. At the core of the design are key components such as the **Arduino UNO microcontroller**, **NodeMCU Wi-Fi module**, and a **solar panel** setup.

The **solar panel** generates DC voltage, which is used to charge a **12V lead-acid battery**. This 12V power is then **boosted to 48V** using an **XL6009 DC-DC boost converter**. The boosted voltage is sufficient to charge a **48V battery** typically used in e-bike or electric scooter applications.

During the charging process, the **Arduino UNO** continuously monitors two key parameters: **battery voltage** and **battery temperature**. These values are displayed in real-time on a **16x2 LCD module**. Based on this data, the system makes intelligent decisions:

- If the **battery temperature** is **below the predefined threshold**, the **relay** is activated, allowing the **boosted voltage** to reach the 48V battery and begin charging.
- If the **temperature crosses the safe threshold**, a **buzzer alert** is triggered, the **relay is deactivated**, and the charging process is **immediately stopped** to protect the battery from thermal damage.

For remote monitoring, the system is integrated with **IoT functionality** using a **NodeMCU module** and the **Blynk cloud server**. This allows voltage and temperature data to be displayed on a smartphone from anywhere in the world, providing convenience and flexibility to users.

In addition to IoT, a **GSM module (SIM800)** is included to send **SMS alerts** to a preconfigured mobile number if any parameter exceeds the threshold limits, serving as an additional layer of safety. The **logical control** of the system is programmed using **Embedded C language**, and code is uploaded to the microcontroller using the **Arduino IDE**. Input components such as sensors are connected to **analog pins**, while output components like the **relay**, **GSM**, **NodeMCU**, and **buzzer** are connected to **digital I/O pins**. The **LCD module** is interfaced using digital pins 2 to 7.

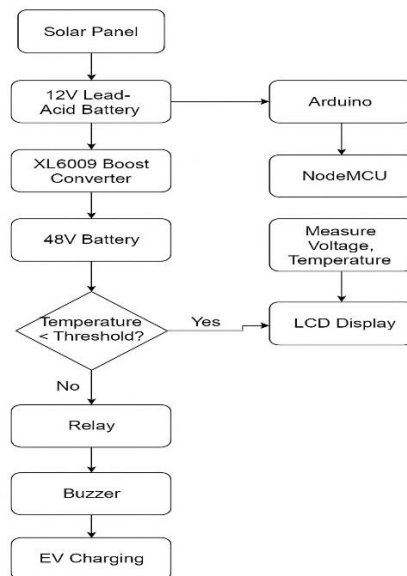


Fig.4.3 Flowchart

One of the key advantages of this system is its ability to **remotely monitor the charging status**, eliminating the need to physically inspect the charger which is often placed outdoors near vehicle parking areas. This feature is particularly beneficial for ensuring user safety and comfort, especially during adverse weather or at night.

4.5 Code Implementation

4.5.1 Introduction

This section presents the software logic implemented in the system using Embedded C and NodeMCU's firmware. The Arduino UNO handles all sensing, controlling, and alerting functions, while the NodeMCU ESP8266 module ensures real-time IoT data transmission using the Blynk platform. Together, these two components enable smart, safe, and connected operation of the solar-based e-bike charging system.

Two sets of code were developed:

- One for the **Arduino UNO**, which also controls the **GSM SIM800 module**, the LCD display, sensors, relay, and buzzer.
- The second for the **NodeMCU**, which reads data from the Arduino and pushes it to the **Blynk Cloud** for wireless monitoring.

4.5.2 Arduino and GSM Code Implementation

This section explains the code written in the Arduino IDE to handle all hardware control, data acquisition from sensors, and communication via GSM and serial output to the NodeMCU. At the beginning of the code, necessary libraries such as LiquidCrystal, DallasTemperature, OneWire, and SoftwareSerial are included to facilitate interaction with the LCD, temperature sensor, and GSM module.

The LCD is initialized using digital pins 2 to 7. The temperature sensor is interfaced using the OneWire protocol on analog pin A0, and the voltage sensor is connected to analog pin A1. The relay is controlled via analog pin A2, and a piezo buzzer is connected to digital pin 13. The GSM module is interfaced through the SoftwareSerial library using pins 8 and 9.

The setup() function initializes serial communication, sensor modules, LCD display, and sets the default states of the relay and buzzer. It also displays welcome messages on the LCD to indicate system initialization. Once the setup is complete, the relay is activated to allow the charging circuit to remain on under normal conditions. The loop() function simply calls the check() function repeatedly, which encapsulates the core logic of the system. Inside check(), two main functions are called:

```

#include<SoftwareSerial.h>
SoftwareSerial myserial(8,9); // GSM

#include<LiquidCrystal.h>
LiquidCrystal lcd(6, 7, 2, 3, 4, 5); // sets the interfacing pins

int current=0, volt=0, temp=0, count=0;
char v;
int a=0, b=0, c=0;

String finalCode;
String a1,b1,c1,d1,e1;

#include <OneWire.h>
#include <DallasTemperature.h>

#define ONE_WIRE_BUS A0
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

/*****VOLTAGE SENSOR VARIABLES*****/

float vout = 0.0;
float vin = 0.0;
float R1 = 200000.0; // resistance of R1 (100K)
float R2 = 10000.0; // resistance of R2 (10K)

/*****/

void setup()
{
  Serial.begin(9600); // GSM
  myserial.begin(9600); // NODEMCU
  sensors.begin();

  pinMode(A2, OUTPUT); // RELAY
  pinMode(13, OUTPUT); // BUZZER

  digitalWrite(A2,LOW); // RELAY OFF
  digitalWrite(13,LOW); // BUZZER OFF

  lcd.begin(16, 2);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("SOLAR BASED 48V");
  lcd.setCursor(0,1);
  lcd.print("EBIKE CHARGER ");
  delay(1000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("USING IOT & GSM");
  lcd.setCursor(0,1);
  lcd.print("TECHNOLOGY ");
  delay(1000);
  digitalWrite(A2,HIGH);
}

void loop()
{
  check();
}

}

void check()
{
  getvoltage(); // GET OUTPUT VOLTAGE OF TRANSFORMER
  gettemperature(); // GET TEMPERATURE OF TRANSFORMER

  // DISPLAY CURRENT, TEMP AND VOLTAGE VALUSE

  lcd.begin(16, 2);
  delay(10);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("BATT VOLT - ");
  lcd.print(volt);

  lcd.setCursor(0,1);
  lcd.print("BATT TEMP - ");
  lcd.print(temp);
  senddata();

  delay(1000);

  // CHECK VOLTAGE CURRENT AND TEMPERATURE , IF ANY PARAMETER INCREASED

  if(temp>39) // TRANSFORMER OVER TEMPERATURE
  {
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("OVER TEMPERATURE");
    lcd.setCursor(0,1);
    lcd.print("CHARGING OFF ");
    // TRIP TRANSFORMER AND SEND SMS
    digitalWrite(A2,LOW); // RELAY OFF
    digitalWrite(13,HIGH); // BUZZER OFF

    delay(1000);
    sendmessage();
    delay(100);
    lcd.begin(16, 2);

    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("OVER TEMPERATURE");
    lcd.setCursor(0,1);
    lcd.print("CHARGING SW. OFF");
    delay(1000);
    senddata2();

    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("BATT VOLT - ");
    lcd.print(volt);

    lcd.setCursor(0,1);
    lcd.print("BATT TEMP - ");
    lcd.print(temp);
  }
}

```

- `getvoltage()`: Reads the battery voltage through the voltage divider circuit using analog pin A1. The actual input voltage is calculated using the formula: $V_{in} = V_{out} \cdot (R1 + R2 / R2)$ where $R1 = 200k\Omega$ and $R2 = 10k\Omega$. A minor calibration offset is also added to fine-tune the measured voltage.
- `gettemperature()`: Reads the temperature using the DS18B20 sensor via the DallasTemperature library. It requests the temperature value and stores it in the variable `temp`.

After reading these values, the LCD is updated with the current battery voltage and temperature. These values are also prepared and sent to the NodeMCU over the serial interface using a formatted string (e.g., "temp@volt*"), which allows real-time monitoring via the Blynk Cloud. The safety mechanism is implemented by comparing the temperature against a threshold value (39°C). If the temperature exceeds this limit, the system automatically performs the following actions:

- Turns off the relay to stop battery charging
- Activates the buzzer to alert users locally
- Displays "Over Temperature" and "Charging Off" on the LCD
- Sends an emergency SMS via the GSM module to a predefined number using AT commands

```
void getvoltage() // READ VOLTAGE
{
  b = analogRead(A1); // VOLTAGE SENSOR
  vout = (b * 5.0) / 1024.0;
  vin = vout / (R2/(R1+R2));
  if (vin<0.00)
  {
    vin=0.0;
  }
  vin=vin+1;
  volt=vin;
  if(volt<=42)
  volt=0;
}

void gettemperature() // READ TEMPERATURE
{
  sensors.requestTemperatures();
  temp = sensors.getTempCByIndex(0);
}

void sendmessage()
{
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("SENDING MESSAGE");
  myserial.println("AT+CMGF=1");
  delay(1000);
  myserial.println("AT+CMGS=\"+919989769753\""); // ENTER REPLY NUMBER
  delay(1000);
  myserial.print("High Alert... Over Temperature - Battery Charging STOP");
  delay(100);
  myserial.println((char)26); // CTRL+Z
  lcd.setCursor(0, 1);
  lcd.print("MESSAGE SENT");
  delay(1000);
}

void senddata2()
{
  a1=String(temp); // TEMP
  b1=String(volt); // VOLT
  finalCode = a1 + "@" + b1 + "*";
  Serial.print(finalCode);
}

void senddata()
{
  if(count == 5)
  {
    a1=String(temp); // TEMP
    b1=String(volt); // VOLT
    finalCode = a1 + "@" + b1 + "*";
    Serial.print(finalCode);
    count = 0;
  }
  count++;
}
```

The code also contains utility functions for communication:

- `sendmessage()`: Uses AT commands to send an SMS alert through the SIM800 GSM module. The message notifies the user of overheating and shutdown of the charging circuit.
- `senddata()` and `senddata2()`: These functions prepare the sensor data in a specific format and send it over the serial interface. This data is captured by the NodeMCU, which is responsible for forwarding it to the **Blynk Cloud**. `senddata()` works on a timed loop, whereas `senddata2()` is triggered during fault conditions.

Role of the Code in the System

This single integrated code plays a central role in the system's automation and safety:

- It collects critical sensor data (temperature and voltage) and processes it in real-time.
- Ensures intelligent charging: charging only occurs if the temperature is within safe limits.
- Uses GSM for fault alerts, ensuring communication even in remote areas with no Wi-Fi.
- Provides real-time display via LCD and remote monitoring via IoT cloud integration.
- Controls key output devices such as the relay (for enabling/disabling charging) and the buzzer (for local alerts).

The logic has been written efficiently using modular functions, making it easy to debug, modify thresholds, or expand to additional sensors or actuators.

4.5.3 NodeMCU Code for IoT Communication

In this project, the NodeMCU ESP8266 is used to enable wireless communication between the Arduino-based control system and the Blynk cloud server. The purpose of this integration is to transmit real-time voltage and temperature data of the e-bike battery to a mobile application so users can monitor the system remotely from anywhere in the world. The NodeMCU code is written in Embedded C using the Arduino IDE.

The NodeMCU establishes a connection to a Wi-Fi network using the `ESP8266WiFi.h` library and sends HTTP GET requests to the Blynk Cloud API using the `ESP8266HTTPClient.h` library. The connection is initiated by providing the SSID and password of the hotspot or Wi-Fi network within the `setup()` function. Once the Wi-Fi connection is successful, the microcontroller begins listening for serial data from the Arduino.

The Arduino sends data in the format `temperature@voltage*`, where:

- The @ symbol separates temperature from voltage,
- The * character marks the end of the message.

The NodeMCU receives this data via the serial interface and parses the incoming string into two variables: one for temperature and one for voltage. Once the values are extracted, they are transmitted to the Blynk cloud using two HTTP GET requests. The data is posted to virtual pins V0 (temperature) and V1 (voltage), which are configured in the Blynk mobile application to display real-time sensor values.

The code implements delays to ensure successful packet delivery and uses the latest method syntax compatible with updated versions of the ESP8266WiFi and HTTP libraries. The WiFiClient client object is created to support the HTTP connection used in the upload process.

The screenshots below illustrate the full NodeMCU code, including Wi-Fi initialization, serial parsing, and cloud communication.

```

1: #include <ESP8266WiFi.h>
   #include <ESP8266HTTPClient.h>

   HTTPClient http;
   WiFiClient client; // NEW LINE ADDED FOR NEW VERSION OF LIBRARY

   int a,b,c,d,e;
   int count;
   char f; String data;
   void setup()
   {
       WiFi.disconnect();
       delay(2000);

       WiFi.begin("project","12345678"); // HOTSPOT ID AND PASSWORD
       while (!WiFi.status() == WL_CONNECTED) {
           delay(300);
       }

       Serial.begin(9600);
       Serial.println("Wifi- Connected");
   }

   void loop()
   {
       while(Serial.available())
       {
           delay(10);
           f = Serial.read();
           if (f == '@')
           {
               a = data.toInt();
               data="";
           }
           else if (f == '*')
           {
               b = data.toInt();
               break;
           }
           else
           {
               data +=f;
           }
       }

       if (data.length() > 0)
       {
           Serial.println(a); // TEMP
           Serial.println(b); // VOLTAGE

           Serial.println("");
           upload();
           data="";
       }
   }

   void upload()
   {
       String link="http://blynk.cloud/external/api/update?token=ttQgPN09973oQ8";
       link=String(link+a);
       http.begin(client,link); // MODIFY SYNTAX FOR NEW VERSION OF LIBRARY ESP
       http.GET();
       http.end();
       delay(50);

       String link1="http://blynk.cloud/external/api/update?token=ttQgPN09973oQ8";
       link1=String(link1+b);
       http.begin(client,link1); // MODIFY SYNTAX FOR NEW VERSION OF LIBRARY ESP
       http.GET();
       http.end();
       delay(50);
   }

```

This code ensures that battery status information can be accessed in real-time by the user without physically inspecting the charger. This feature improves safety, convenience, and reliability of the entire charging system, fulfilling one of the core objectives of the IoT-enabled project.

4.6 Advantages of the Proposed System

The proposed solar-powered 48V e-bike charging system integrated with IoT-based protection and monitoring offers several benefits over conventional charging solutions. These advantages are both **technical** and **user-centric**, ensuring enhanced reliability, safety, and sustainability.

- **Renewable Energy Utilization:** The system is powered by solar energy, promoting the use of clean and sustainable energy for electric vehicle charging. This reduces dependency on grid power and contributes to a greener environment.
- **Battery Health Monitoring:** Real-time monitoring of voltage and temperature helps extend battery life by preventing overcharging, deep discharging, and overheating. These parameters are crucial in maintaining the overall health of the battery.
- **IoT-Based Remote Monitoring:** The integration of the NodeMCU and Blynk IoT platform allows users to track battery status and charging conditions remotely through a smartphone application. This provides flexibility and peace of mind for users.
- **Safety Features with Alerts:** The inclusion of a buzzer and GSM module ensures immediate local and remote alerts in case of over-temperature or voltage fluctuations. This enhances system safety and protects components from damage.
- **Cost-Efficient and User-Friendly:** The design leverages affordable components such as the Arduino UNO, GSM module, and NodeMCU, making it a cost-effective solution. The LCD display provides local updates, and the system is easy to maintain.
- **Automation with Manual Override:** The charging process is fully automated based on real-time sensor readings. Additionally, manual switches allow users to control the system locally if needed.
- **Scalable and Flexible Design:** The modular nature of the design makes it easy to scale for higher power levels or add more sensors and features as per future requirements.
- **Offline and Online Communication:** With both GSM and Wi-Fi-based monitoring, the system remains functional even if one communication mode fails, ensuring high availability and redundancy.

4.7 Limitations of the System

Despite the successful implementation and functioning of the solar-based 48V e-bike charger with IoT-enabled protection, the system is subject to a few limitations:

- **Dependence on Solar Availability:** The charging efficiency is highly dependent on the availability of sunlight. On cloudy or rainy days, the energy output from the solar panel is significantly reduced.
- **Limited Battery Backup:** The 12V intermediate battery has a limited capacity, which restricts the duration of charging when sunlight is not available.
- **Sensor Accuracy:** Low-cost analog sensors may exhibit minor fluctuations or inaccuracies, affecting decision-making logic.
- **IoT Connectivity Issues:** If Wi-Fi or GSM signals are weak or disrupted, real-time monitoring or alert messages may be delayed or missed.
- **Manual Setup:** The system requires manual calibration of threshold values (voltage and temperature), which might vary depending on battery specifications.
- **Single EV Battery Support:** The current system is designed to charge one e-bike battery at a time. It would need scaling up for multi-battery or commercial use cases.
- These limitations can be addressed in future versions with better hardware, dynamic tracking algorithms, and larger power capacity systems.

4.8 Challenges Faced During Implementation

During the development and implementation of the system, several technical and practical challenges were encountered:

- **Voltage Instability from Solar Panel:** Due to natural fluctuations in sunlight intensity, the panel's voltage output varied frequently. This required fine-tuning of the voltage regulation and filtering stages.
- **Sensor Calibration:** Getting accurate readings from the voltage and temperature sensors was a challenge initially. The team had to adjust the analog conversion logic and test across different load conditions.
- **Boost Converter Tuning:** Achieving a stable 48V output from a 12V input involved several iterations in adjusting the XL6009 module and ensuring it didn't overheat.
- **IoT Integration Delays:** The Blynk platform and NodeMCU experienced occasional lags due to internet issues, which temporarily hindered live monitoring.

- **Power Distribution Conflicts:** Sharing power between the Arduino, NodeMCU, and other components caused brownouts in some tests. This was resolved by using separate voltage regulators and capacitors.
- **Coding and Serial Conflicts:** Since both the GSM and NodeMCU used serial communication, managing serial port switching without interference was a challenging task.

Each challenge strengthened the robustness of the system and helped develop a deeper understanding of hardware-software integration.

4.9 Summary of Chapter

This chapter detailed the complete design and implementation process of the solar-powered 48V e-bike charger with IoT-enabled protection. The chapter began with the system's block and circuit diagrams, highlighting the functional flow from solar input to e-bike battery charging. It explained the working principle and methodology behind each component, including sensors, microcontrollers, and communication modules.

Software implementation using Embedded C, Arduino IDE, and the Blynk IoT platform was elaborated upon, with detailed insights into the embedded logic and serial communication. The system's advantages in renewable energy usage, safety, and remote monitoring were balanced with a discussion of its current limitations and the technical challenges faced during development.

Overall, this chapter demonstrated how different technologies and components were seamlessly integrated to deliver a practical, efficient, and intelligent EV charging solution.

CHAPTER 5

CHAPTER 5

Prototype And Results

5.1 Introduction

This chapter highlights the experimental validation and functional output of the developed system titled "**Solar-Based 48V E-Bike Charger with Protection Using IoT.**" The main objective of this project was to design a sustainable and safe charging system for electric bikes using solar energy, integrated with modern technologies like IoT and GSM for intelligent monitoring and control.

The implemented system was tested under real-time conditions, and the output was observed through local indicators (LCD and buzzer) as well as remotely via the Blynk IoT platform. The key performance parameters monitored include battery voltage, charging status, and battery temperature, which are crucial for ensuring the safety and longevity of the battery.

This chapter documents how the system behaves under different input conditions, particularly during over-voltage or over-temperature scenarios. The results obtained confirm that the system works efficiently, providing safe and intelligent charging while also enabling remote supervision. The inclusion of visual outputs such as prototype images, LCD screenshots, Blynk app interface, and serial monitor logs further validates the functionality of the system.

5.2 Working Prototype

The physical prototype integrates all hardware components, including the solar panel, boost converter, 12V battery, sensors, Arduino, NodeMCU, relay module, GSM modem, LCD, and buzzer. All modules are mounted and interconnected as per the circuit design.

Key Features Demonstrated:

- Solar charging of the 12V battery and boosting to 48V.
- Real-time voltage and temperature monitoring.
- Automatic relay control for safe battery charging.
- Audible buzzer alerts during fault conditions.
- Remote monitoring via the Blynk IoT dashboard.
- SMS alerts using the GSM module during over-temperature conditions.

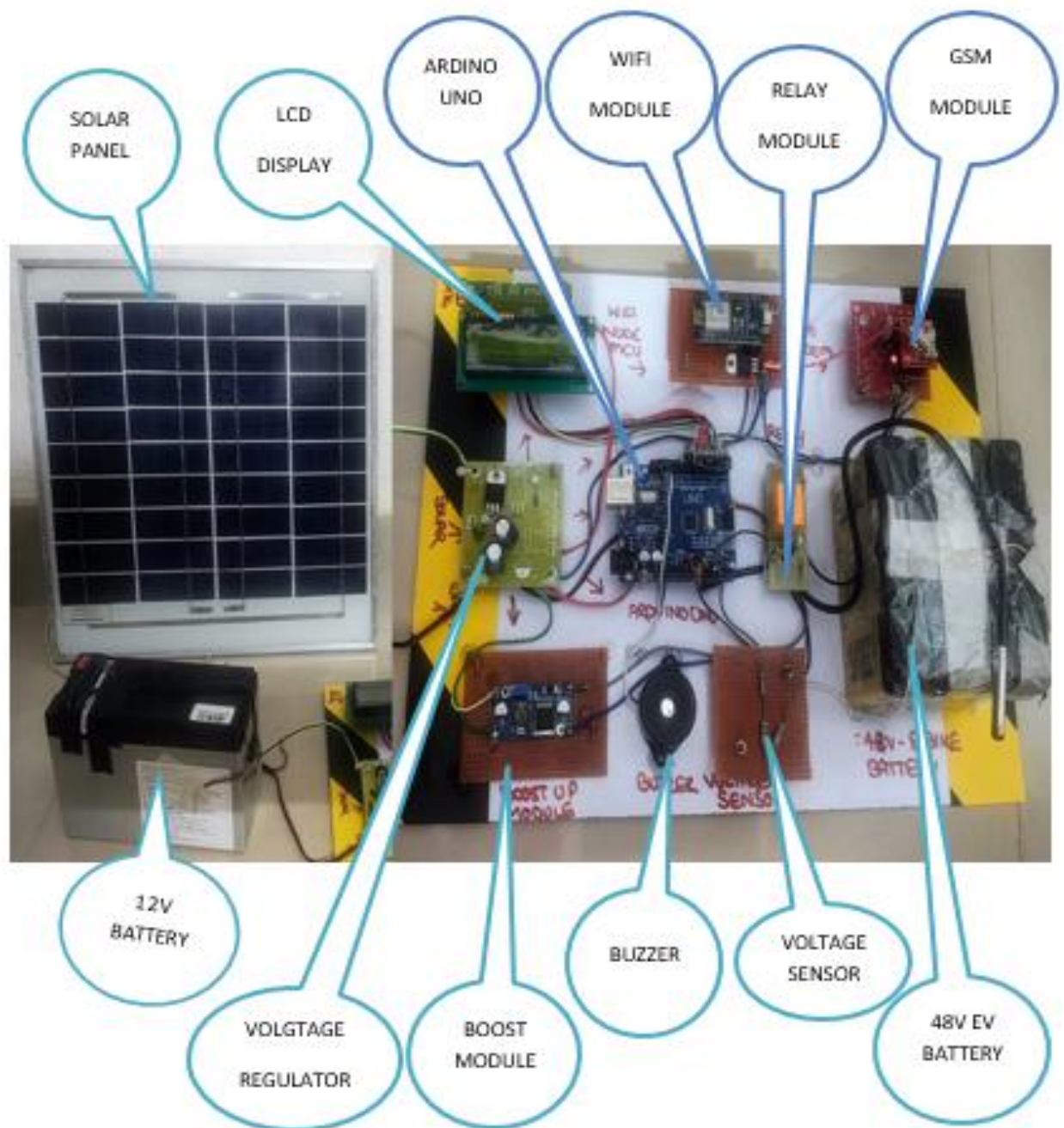


Fig.5.1 Prototype

5.3 Output Pictures and Observations

This section presents photographic evidence of the working prototype during various stages of its operation. Each image captured here reflects the actual outputs displayed on the 16x2 LCD module, mounted on the hardware setup. These pictures are taken in real-time during the prototype's functioning and offer a clear understanding of how the system behaves under different operating conditions. The display shows critical information like power status, battery voltage, temperature readings, and system alerts, all of which are essential for the safe and efficient operation of an electric vehicle charging and protection system. These outputs also

validate the integration of IoT and GSM technologies in real-time monitoring and data transmission. Overall, the images serve as proof-of-concept for the design and successful implementation of the proposed solution.

Fig. 5.2 & Fig. 5.3 – System Initialization and Battery Monitoring



In **Fig. 5.2**, the system is powered through the 12V battery, and the LCD display confirms startup with the message “*USING IOT & GSM TECHNOLOGY*”. This output indicates that the system has successfully received power and entered its initialization stage. The display serves as a visual indicator that the embedded logic and microcontroller setup have started running, marking the readiness of all interfaced modules, including sensors, GSM, and NodeMCU.

Moving to **Fig. 5.3**, the connection of the main 48V e-bike battery is confirmed through the LCD, which now updates to reflect live values for both *battery voltage* and *temperature*. This output demonstrates the real-time functioning of the voltage and temperature sensors and verifies that the data is correctly processed and displayed. The integration of sensing and control elements is evident from this image, confirming that the system is now actively monitoring and ready to charge the EV battery under safe conditions.

Fig. 5.4 – Battery Monitoring via Blynk Application



In addition to the local LCD output, the system incorporates wireless monitoring through the Blynk IoT mobile application. **Fig. 5.4** displays real-time battery parameters—specifically, a voltage of 48V and a temperature of 31°C—as seen on a mobile device. This is made possible through the integration of a NodeMCU Wi-Fi module, which transmits sensor data to the Blynk cloud. The remote interface enhances user convenience,

allowing the charging process to be monitored from anywhere. This feature improves safety, promotes energy efficiency, and reflects the modern shift toward smart, connected EV charging solutions.

Fig. 5.5 & Fig. 5.6 – Overheating Protection Mechanism

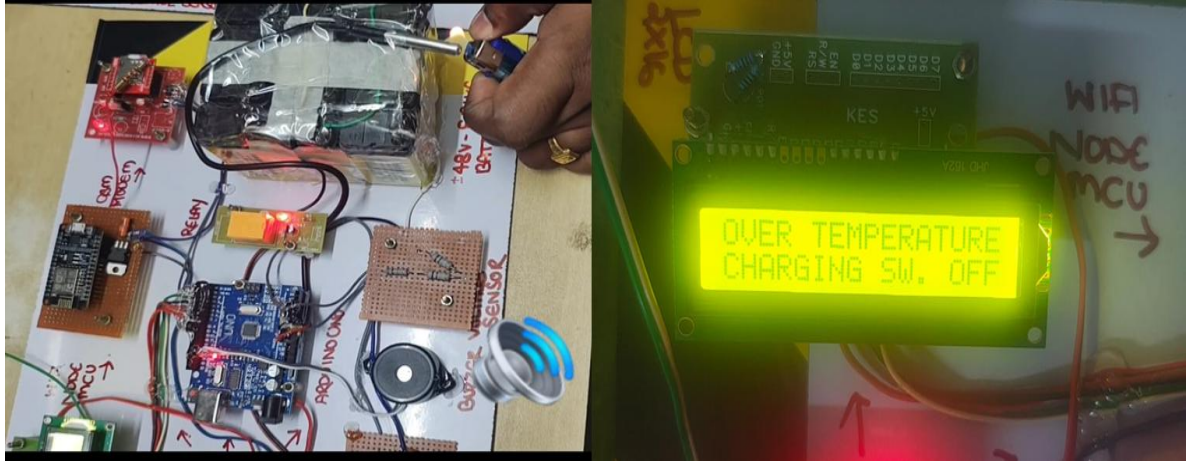
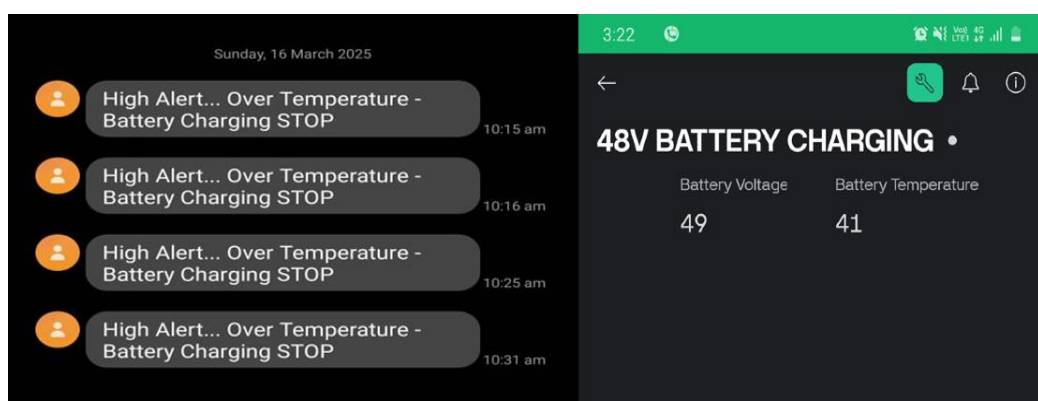


Fig. 5.5, the system monitors the battery temperature using a temperature sensor. When heat is applied to the battery, the sensor detects the rising temperature and sends a signal to the Arduino Uno. This triggers the buzzer, providing an immediate audible alert to indicate overheating. The image also highlights key components such as the GSM module and NodeMCU, which support system monitoring and control.

In Fig. 5.6, the system has detected an over-temperature condition, and the LCD display confirms an automatic shutdown of the charging process with the message: **"OVER TEMPERATURE CHARGING SW. OFF."** The relay immediately disconnects the power supply to prevent potential overheating hazards. Simultaneously, the buzzer provides an audible warning to alert the user. This automated protection mechanism ensures battery longevity, prevents thermal runaway, and enhances the overall safety of the EV charging system.

Fig. 5.7 & Fig. 5.8– Remote Monitoring and Safety Alerts



In Fig. 5.7, the system implements an instant safety alert mechanism using the GSM module. When the battery temperature exceeds the predefined safety limit, an automated SMS notification is sent to the user. The repeated alerts emphasize the critical nature of overheating, ensuring that immediate action can be taken to prevent damage. Additionally, the system automatically stops the charging process to prevent further temperature rise, demonstrating an integrated safety feature. This real-time alert system enhances user awareness and allows remote monitoring of battery health, ensuring a more secure and reliable charging experience.

In Fig. 5.8, the Blynk application provides real-time battery monitoring, displaying key parameters such as voltage and temperature. The interface shows that the battery voltage is at 49V while the temperature has risen to 41°C. This data is transmitted wirelessly using a NodeMCU Wi-Fi module, allowing users to monitor the charging status remotely. By offering a live feed of battery conditions, the system enhances convenience and ensures safe charging operations. This IoT-based monitoring approach helps users take timely action in case of overheating, making the system more efficient and user-friendly.

CHAPTER 6

CHAPTER 6

Conclusion And Future Scope

6.1 Conclusion

The aim of this project was to develop a smart and sustainable **solar-based charging system for electric vehicles (EVs)** with integrated protection features using IoT and GSM technologies. The system effectively utilizes solar energy to charge a 48V e-bike battery and ensures real-time monitoring and intelligent control through embedded hardware components like Arduino and NodeMCU.

A key highlight of this system is the mobile-based monitoring enabled through the **Blynk application**, which allows users to remotely track battery voltage and temperature in real-time. The integration of **GSM technology** further enhances safety by sending SMS alerts when sensor readings exceed defined thresholds, preventing potential damage to the battery and charging system.

The use of simulation tools and cloud-based platforms has allowed for better analysis of system behavior under various operating conditions. The **IoT-based framework**, when combined with cloud storage and mobile access, proves to be a powerful solution for both energy management and user engagement. This project demonstrates how embedded systems and renewable energy can be seamlessly combined to provide a user-friendly, reliable, and eco-conscious EV charging solution.

6.2 Future Scope

As electric vehicle adoption grows and sustainability becomes a priority, this system can be further improved in several ways to make it more reliable, user-friendly, and commercially viable. Below are the possible areas for future development:

- **RFID-Based Access and Payment Integration:**

Incorporating **RFID card authentication** will allow only registered users to access the charging facility. These RFID cards can be linked to **prepaid user accounts** through popular UPI-based payment apps like Google Pay or Paytm. Every time a user initiates a

charge, the system can deduct the corresponding amount from their balance. This eliminates the need for cash payments and enables **automated billing**. It also increases the commercial potential of the system for public and shared EV charging infrastructure.

- **Enhanced Data Analytics and Cloud Dashboard :**

Currently, the system shows only live voltage and temperature data. In the future, data from sensors can be stored on a cloud server and visualized through a **custom web dashboard**. This will enable users to analyze battery performance trends over time and help in **predictive maintenance**. Notifications for upcoming maintenance, daily charge cycles, and efficiency metrics can be integrated. Such data insights will add value for both end users and service providers managing charging stations.

- **Expanded Device Compatibility and Custom App:**

At present, the system uses the Blynk platform for IoT monitoring. In the future, a **custom-developed Android or iOS application** can provide more control, personalized features, and branding. Such an app can include features like **real-time status alerts**, usage history, custom thresholds, and fault logs. The same app can support multiple EV chargers in different locations. Additionally, support for **multi-language and voice control** features can be added to enhance accessibility.

- **Scalability for Higher Voltage EVs:**

The current design is suited for 48V e-bikes, but the system can be modified to support **72V or 96V EVs**, such as electric scooters and cars. This requires upgrading the boost converter, sensors, and relay to handle higher voltages and currents. The battery protection algorithm must also be adapted for higher capacity lithium-ion batteries. With these upgrades, the system could be implemented as a **community or city-level EV charging station**. It also paves the way for integrating with solar microgrids.

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